

Two-Timescale Secure Transmission for Movable Antenna-Enabled Multiuser Systems With Statistical Eavesdropper CSI

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Abstract—Movable antennas (MAs) can reshape the spatial correlation between the channels of legitimate users (LUs) and eavesdroppers, providing a new degree of freedom for physical layer security. However, existing MA-aided secure designs mostly adjust the antenna positions based on instantaneous channel state information (CSI), which is hindered by the antenna movement delay, the overhead of acquiring the CSI over the entire movable region, and the unavailability of the instantaneous CSI of passive eavesdroppers. To address these issues, we propose a two-timescale secure transmission framework for an MA-enabled multiuser downlink system with a passive eavesdropper (Eve), where the antenna positions and the long-term power allocation among the confidential signals and artificial noise (AN) are optimized based on the statistical CSI, while zero-forcing (ZF) precoding and null-space AN are designed based on the instantaneous CSI of the LUs only. By exploiting the conditional independence between the information leakage and the AN received by Eve, we derive a closed-form approximation of the ergodic wiretap rate, which reveals that the antenna positions affect the ergodic secrecy sum rate (ESSR) only through three geometric quantities characterizing the decorrelation among the LUs, the information leakage, and the AN received by Eve. Then, an alternating optimization algorithm is developed to maximize the ESSR, where the power allocation admits a secrecy water-filling structure and the antenna positions are updated element-wise via the majorization-minimization method. Numerical results show that the proposed design significantly outperforms fixed-position antenna arrays, enhances the secrecy performance even for a single LU, and reduces the required AN power.

Index Terms—Movable antenna, physical layer security, artificial noise, two-timescale design, statistical CSI, ergodic secrecy rate.

I. INTRODUCTION

PHYSICAL layer security (PLS) exploits the randomness and the spatial characteristics of wireless channels to protect confidential messages against eavesdropping, which is a promising complement to upper-layer cryptography [1]–[3]. In multi-antenna systems, secure precoding and artificial noise (AN) are two main enablers of PLS [4], [5]. However, their performance is fundamentally limited by how well the legitimate users (LUs) and the eavesdroppers can be spatially separated. When an eavesdropper (Eve) is located in a direction close to that of an LU and the line-of-sight (LoS) propagation is dominant, the channels of the LU and Eve are highly correlated. In this case, a conventional fixed-position antenna (FPA) array can hardly steer the information-bearing beams away from Eve while maintaining the beamforming gain at the LU.

Movable antennas (MAs), also known as fluid antennas, have recently emerged as a promising technology to overcome this limitation [6]–[8]. By adjusting the antenna positions within a given region via flexible cables and drivers, MAs reshape the channel correlation among different receivers without additional radio-frequency (RF) chains [9]–[11]. This capability is particularly appealing for PLS, since the antenna positions can be designed to decorrelate the channels of the LUs from that of Eve. Motivated by this, MA-aided secure transmission has attracted growing research interest. For example, the authors of [12] jointly optimized the transmit beamforming and the positions of an MA array to maximize the secrecy rate. MA-aided secure transmission without the instantaneous channel state information (CSI) of the eavesdroppers was investigated in [13], where the statistical CSI of the eavesdroppers was exploited. More recently, the ergodic secrecy rate of multiple-input multiple-output multiple-eavesdropper-antenna (MIMOME) MA systems with statistical eavesdropper CSI was studied in [14], and a sensing-assisted secure design was developed in [15], where the MA positions were optimized to enhance both the estimation of the direction of Eve and the secrecy rate.

Despite these advances, most of the above works optimize the antenna positions based on the instantaneous CSI of at least the LUs, which requires the antennas to be repositioned in each channel coherence interval (CI) under fast fading. Such single-timescale designs are hardly practical for the following three reasons. First, the mechanical movement of antennas incurs non-negligible delay and energy consumption, which can hardly keep pace with the channel variations. Second, optimizing the antenna positions based on instantaneous CSI requires the CSI over the entire movable region. Under Rician fading, the non-LoS (NLoS) components at different positions cannot be characterized by a few path parameters, and thus acquiring them requires the antennas to traverse the region [16], [17]. Third, a passive Eve does not transmit, and thus its instantaneous CSI is generally unavailable at the transmitter.

To reduce the implementation overhead, two-timescale designs have been proposed for MA-enabled systems [16], [17], following the two-timescale paradigm developed for intelligent reflecting surfaces [18]. In these designs, the antenna positions are optimized based on the statistical CSI in the large timescale, whereas the transmit or receive beamforming is designed based on the instantaneous CSI of the resulting effective channels in the small timescale. However, these works focused on the ergodic sum rate, and the two-timescale design of MA-

enabled secure transmission has not been investigated yet. In particular, how to exploit the statistical CSI of a passive Eve to position the MAs, and how the MAs interact with AN in the two-timescale framework, remain open problems.

Extending the two-timescale MA design to secure transmission is not straightforward, since the role of the antenna positions fundamentally changes in the presence of Eve. First, since the instantaneous CSI of Eve is unknown, the achievable ergodic secrecy rate involves the ergodic wiretap rate of Eve, which is determined by the random precoding directions and AN subspace induced by the instantaneous CSI of the LUs. The existing ergodic-rate approximations for zero-forcing (ZF) precoding [16], [17], [19] characterize only the signal power received by the intended users and cannot be applied to Eve. Second, in the rate-oriented design, the antenna positions improve the ergodic rate only by decorrelating the channels of the users, and thus bring no gain for a single user [16]. In contrast, in secure transmission, the antenna positions also determine the information leakage to Eve and the AN received by Eve. Third, the power allocation between the confidential signals and the AN is coupled with the antenna positions, since reducing the channel correlation between the LUs and Eve changes the amount of AN needed.

Motivated by the above, this paper investigates the two-timescale design of MA-enabled secure multiuser downlink transmission in the presence of a passive Eve. The main contributions are summarized as follows.

- We propose a two-timescale secure transmission framework for MA-enabled multiuser systems. In the large timescale, the antenna positions and the long-term power allocation among the confidential signals and AN are optimized based on the statistical CSI of the LUs and Eve. In the small timescale, ZF precoding and null-space AN are designed based on the instantaneous CSI of the LUs, which requires neither the instantaneous CSI of Eve nor frequent antenna movement.
- We derive a closed-form approximation of the ergodic wiretap rate of Eve. Specifically, we prove that the information leakage and the AN received by Eve are conditionally independent given the channels of the LUs, and develop first-order approximations of their means that are exact in both the NLoS- and LoS-dominant regimes. Combined with the existing ergodic-rate approximation for ZF precoding, the resulting ESSR depends on the antenna positions only through three geometric quantities, which reveals that the MAs enhance secrecy by decorrelating the LUs, reducing the information leakage, and increasing the AN received by Eve. In particular, different from the rate-oriented design, the MAs enhance the secrecy performance even for a single LU.
- We formulate an ESSR maximization problem by jointly optimizing the antenna positions and the long-term power allocation, and develop an alternating optimization (AO) algorithm to solve it. The power allocation subproblem admits a secrecy water-filling solution combined with a one-dimensional search over the AN power, which switches off the LUs with non-positive secrecy rates. The antenna positions are optimized in an element-wise man-

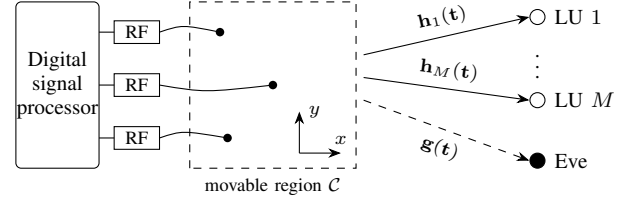


Fig. 1. An MA-enabled multiuser downlink system in the presence of a passive eavesdropper.

ner by the majorization–minimization (MM) method with analytical gradients. The proposed algorithm converges monotonically, and its complexity does not depend on the number of channel realizations.

- Numerical results validate the accuracy of the derived approximation and demonstrate that the proposed design significantly outperforms the FPA-based benchmarks, e.g., by 28.1% and 13.6% over a compact $\lambda/2$ array and a sparse array spanning the movable region, respectively, at a transmit SNR of 20 dB. Moreover, the results show that the MAs reduce the AN power required for secure transmission, and that the long-term per-user power allocation is essential for LUs with heterogeneous large-scale fading.

The remainder of this paper is organized as follows. Section II presents the system model and formulates the two-timescale ESSR maximization problem. Section III specifies the ZF precoding and null-space AN and derives the approximate ESSR. Section IV develops the two-timescale optimization algorithm. Section V provides numerical results, and Section VI concludes this paper.

Notations: Scalars, vectors, and matrices are denoted by italic, bold lower-case, and bold upper-case letters, respectively. $(\cdot)^T$, $(\cdot)^*$, and $(\cdot)^H$ denote the transpose, conjugate, and conjugate transpose, respectively. $|\cdot|$, $\|\cdot\|$, and $\text{tr}(\cdot)$ denote the modulus of a complex scalar, the Euclidean norm of a vector, and the trace of a square matrix, respectively. $[\mathbf{x}]_i$ and $[\mathbf{X}]_{i,j}$ denote the i -th entry of $\mathbf{x}$ and the (i,j) -th entry of $\mathbf{X}$, respectively, and $\text{diag}(x_1, \dots, x_M)$ denotes a diagonal matrix with diagonal entries $x_1, \dots, x_M$. For a diagonal matrix $\mathbf{X}$ with nonnegative entries, $\mathbf{X}^{1/2}$ takes the square root of each entry. $\mathbf{I}_N$ denotes the $N \times N$ identity matrix, and $\mathbf{e}_m$ denotes the m -th column of an identity matrix whose dimension is clear from the context. $\mathbb{C}^{a \times b}$ and $\mathbb{R}^{a \times b}$ denote the spaces of $a \times b$ complex and real matrices, respectively. $\mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes the circularly symmetric complex Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. $\mathbf{X} \succeq \mathbf{0}$ ($\mathbf{X} \succ \mathbf{0}$) means that $\mathbf{X}$ is positive semidefinite (definite). $j \triangleq \sqrt{-1}$ denotes the imaginary unit, $\mathbb{E}\{\cdot\}$ the expectation, $\text{Re}\{\cdot\}$ the real part, $[x]^+ \triangleq \max\{x, 0\}$, ∇f the gradient of f , $\triangleq$ a definition, and $\mathcal{O}(\cdot)$ the big-O notation.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. MA-Enabled Secure Multiuser Downlink System

As shown in Fig. 1, we consider the downlink of a multiuser system, where a base station (BS) equipped with N MAs

serves M single-antenna LUs in the presence of a single-antenna passive Eve. The LUs and the MAs are indexed by $m \in \mathcal{M} \triangleq \{1, \dots, M\}$ and $n \in \mathcal{N} \triangleq \{1, \dots, N\}$, respectively, where $N \geq M + 1$ is assumed so that AN can be transmitted in the null space of the LU channels. The MAs are connected to the RF chains via flexible cables, which allows them to move within a two-dimensional (2D) square region $\mathcal{C} \triangleq [-A/2, A/2] \times [-A/2, A/2]$ [6], [7]. The position of the n -th MA is denoted by $\mathbf{t}_n \triangleq [x_n, y_n]^T \in \mathcal{C}$, and the positions of all MAs are collected in $\mathbf{t} \triangleq [\mathbf{t}_1^T, \dots, \mathbf{t}_N^T]^T \in \mathbb{R}^{2N \times 1}$. To avoid the coupling effect between adjacent antennas, any two MAs are separated by at least a minimum distance $D_{\min}$, i.e., $\|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min}, \forall n \neq i$.

Since the size of $\mathcal{C}$ is much smaller than the signal propagation distance, the far-field condition holds between the BS and the receivers. As a result, moving the MAs changes only the phase of the LoS path, whereas the angle of departure (AoD) and the amplitude of the LoS path remain unchanged [7]. Let $\theta_m \in [-\pi/2, \pi/2]$ and $\phi_m \in [-\pi/2, \pi/2]$ denote the elevation and azimuth AoDs of the LoS path from the BS to LU m , respectively, and define the corresponding wave vector $\mathbf{a}_m \triangleq [\cos \theta_m \sin \phi_m, \sin \theta_m]^T$. Then, the propagation distance difference of the LoS path between position $\mathbf{t}_n$ and the origin of $\mathcal{C}$ is $\mathbf{t}_n^T \mathbf{a}_m$, and the LoS component of the channel from the BS to LU m is given by [16]

$$\bar{\mathbf{h}}_m(\mathbf{t}) \triangleq \left[e^{j\frac{2\pi}{\lambda} \mathbf{t}_1^T \mathbf{a}_m}, \dots, e^{j\frac{2\pi}{\lambda} \mathbf{t}_N^T \mathbf{a}_m} \right]^T \in \mathbb{C}^{N \times 1}, \quad (1)$$

where λ is the carrier wavelength. Following [16], [17], the channel from the BS to LU m is modeled as Rician fading, i.e.,

$$\mathbf{h}_m(\mathbf{t}) = \sqrt{\frac{\kappa_m \beta_m}{\kappa_m + 1}} \bar{\mathbf{h}}_m(\mathbf{t}) + \sqrt{\frac{\beta_m}{\kappa_m + 1}} \tilde{\mathbf{h}}_m, \quad (2)$$

where β_m is the large-scale fading coefficient, $\kappa_m \geq 0$ is the Rician factor, and $\tilde{\mathbf{h}}_m \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$ is the NLoS component.¹ Similarly, the channel from the BS to Eve is given by

$$\mathbf{g}(\mathbf{t}) = \sqrt{\frac{\kappa_e \beta_e}{\kappa_e + 1}} \bar{\mathbf{g}}(\mathbf{t}) + \sqrt{\frac{\beta_e}{\kappa_e + 1}} \tilde{\mathbf{g}}, \quad (3)$$

where β_e , κ_e , and $\tilde{\mathbf{g}} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$ are defined similarly to those in (2), and $\bar{\mathbf{g}}(\mathbf{t})$ is given by (1) with $\mathbf{a}_m$ replaced by $\mathbf{a}_e \triangleq [\cos \theta_e \sin \phi_e, \sin \theta_e]^T$, with θ_e and ϕ_e being the elevation and azimuth AoDs of Eve, respectively. The NLoS components $\tilde{\mathbf{h}}_1, \dots, \tilde{\mathbf{h}}_M$ and $\tilde{\mathbf{g}}$ are mutually independent. For convenience, we define $\mathbf{H}(\mathbf{t}) \triangleq [\mathbf{h}_1(\mathbf{t}), \dots, \mathbf{h}_M(\mathbf{t})] \in \mathbb{C}^{N \times M}$, $\tilde{\mathbf{H}}(\mathbf{t}) \triangleq [\tilde{\mathbf{h}}_1(\mathbf{t}), \dots, \tilde{\mathbf{h}}_M(\mathbf{t})]$, $\tilde{\mathbf{H}} \triangleq [\tilde{\mathbf{h}}_1, \dots, \tilde{\mathbf{h}}_M]$, and $\mathbf{K} \triangleq \text{diag}(\kappa_1, \dots, \kappa_M)$. Note that the antenna positions affect the channels only through the LoS components $\tilde{\mathbf{H}}(\mathbf{t})$ and $\bar{\mathbf{g}}(\mathbf{t})$.

B. Two-Timescale Transmission Protocol

Under Rician fading, the NLoS components at different antenna positions are random. Acquiring the instantaneous

¹ As in [16], [17], the NLoS components at different antennas are assumed to be independent and identically distributed (i.i.d.), which is reasonable in rich-scattering environments with $D_{\min} \geq \lambda/2$. Spatially correlated NLoS components can be incorporated at the cost of more involved expressions.

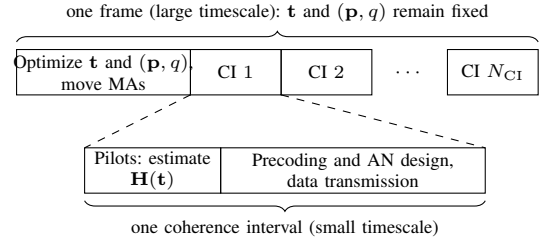


Fig. 2. Illustration of the two-timescale transmission protocol (CI: coherence interval).

CSI over the entire region $\mathcal{C}$ would require the MAs to traverse the region, which incurs prohibitive antenna movement delay and channel training overhead [16], [17]. Moreover, since Eve is passive, the instantaneous CSI of its channel is unavailable at the BS. In contrast, the statistical CSI, i.e., $\{\beta_m, \kappa_m, \theta_m, \phi_m\}_{m \in \mathcal{M}}$ of the LUs and $\{\beta_e, \kappa_e, \theta_e, \phi_e\}$ of Eve, is determined by the large-scale fading and the LoS geometry, which vary much more slowly than the small-scale fading. Following [13], [15], we assume that the statistical CSI of Eve is known at the BS, which can be obtained, e.g., from the prior knowledge of the location of Eve, by sensing the direction of Eve [15], or from the historical signals of Eve if it is an untrusted node of the network.

Accordingly, we adopt the two-timescale transmission protocol illustrated in Fig. 2. The time is divided into frames, each of which consists of $N_{\text{CI}} \gg 1$ CIs over which the statistical CSI remains constant. At the beginning of each frame, the BS optimizes the antenna positions $\mathbf{t}$ and the long-term transmission parameters based on the statistical CSI, and moves the MAs accordingly. Then, with $\mathbf{t}$ fixed, the BS estimates the instantaneous CSI of the LUs, i.e., $\mathbf{H}(\mathbf{t})$, via uplink pilots in each CI by exploiting the channel reciprocity of time-division duplexing, and designs the transmit precoding and AN. For the purpose of exposition, $\mathbf{H}(\mathbf{t})$ is assumed to be perfectly estimated in each CI, whereas the instantaneous CSI of Eve, i.e., $\tilde{\mathbf{g}}$, is never available at the BS. Compared with the instantaneous-CSI-based MA designs, the proposed protocol moves the MAs only once per frame and requires the instantaneous CSI of the LUs only at the current antenna positions, which significantly reduces the antenna movement and channel estimation overhead.

C. Signal Model

In each CI, the BS transmits the confidential symbols $s_m \sim \mathcal{CN}(0, 1)$, $m \in \mathcal{M}$, together with AN, i.e., the transmitted signal is given by

$$\mathbf{x} = \sum_{m=1}^M \mathbf{w}_m s_m + \mathbf{z}, \quad (4)$$

where $\mathbf{w}_m \in \mathbb{C}^{N \times 1}$ is the precoding vector for LU m , and $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{\Sigma})$ is the AN vector with covariance matrix $\mathbf{\Sigma} \succeq \mathbf{0}$, which is independent of the confidential symbols. The symbols $\{s_m\}$ are independent of each other. Under the two-timescale protocol, $\{\mathbf{w}_m\}$ and $\mathbf{\Sigma}$ are designed in each CI based on $\mathbf{H}(\mathbf{t})$, and are thus functions of the instantaneous CSI of the LUs,

which are denoted by $\mathbf{w}_m(\mathbf{H})$ and $\mathbf{\Sigma}(\mathbf{H})$ when necessary. The transmit power of the BS is limited by $\sum_{m=1}^M \|\mathbf{w}_m\|^2 + \text{tr}(\mathbf{\Sigma}) \leq P_{\text{tot}}$.

The signal received at LU m is $y_m = \mathbf{h}_m^H(\mathbf{t})\mathbf{x} + n_m$, where $n_m \sim \mathcal{CN}(0, \sigma_m^2)$ is the additive white Gaussian noise (AWGN). Hence, the signal-to-interference-plus-noise ratio (SINR) of LU m is given by

$$\gamma_m = \frac{|\mathbf{h}_m^H \mathbf{w}_m|^2}{\sum_{i \in \mathcal{M} \setminus \{m\}} |\mathbf{h}_m^H \mathbf{w}_i|^2 + \mathbf{h}_m^H \mathbf{\Sigma} \mathbf{h}_m + \sigma_m^2}, \quad (5)$$

where the dependence of the channels on $\mathbf{t}$ is omitted for brevity. The signal received at Eve is $y_e = \mathbf{g}^H(\mathbf{t})\mathbf{x} + n_e$, where $n_e \sim \mathcal{CN}(0, \sigma_e^2)$ is the AWGN. To guarantee secrecy in the worst case, we assume that Eve is able to cancel the signals of the other LUs when intercepting s_m , e.g., via multiuser detection [20]. Accordingly, the SINR of Eve for intercepting s_m is given by

$$\gamma_{e,m} = \frac{|\mathbf{g}^H \mathbf{w}_m|^2}{\mathbf{g}^H \mathbf{\Sigma} \mathbf{g} + \sigma_e^2}, \quad (6)$$

which upper-bounds the SINR of Eve when the signals of the other LUs are treated as interference.

D. Ergodic Secrecy Rate and Problem Formulation

Since the instantaneous CSI of Eve is unknown, the BS cannot adapt the secrecy coding to the instantaneous channel of Eve. Instead, the confidential message of each LU is encoded over the CIs of a frame, which experience independent realizations of the NLoS components. By coding over these fading realizations, the following ergodic secrecy rate is achievable for LU m [20], [21]:

$$R_m^{\text{sec}} \triangleq [\bar{R}_m - \bar{R}_{e,m}]^+, \quad (7)$$

where

$$\bar{R}_m \triangleq \mathbb{E}\{\log_2(1 + \gamma_m)\}, \quad \bar{R}_{e,m} \triangleq \mathbb{E}\{\log_2(1 + \gamma_{e,m})\} \quad (8)$$

denote the ergodic rate of LU m and the ergodic wiretap rate of Eve for s_m , respectively, and the expectations are taken over the NLoS components $\tilde{\mathbf{H}}$ and $\tilde{\mathbf{g}}$ for given $\mathbf{t}$. Note that R_m^{sec} depends on both the antenna positions $\mathbf{t}$ and the short-timescale design policies $\{\mathbf{w}_m(\cdot)\}$ and $\mathbf{\Sigma}(\cdot)$.

We aim to maximize the ergodic secrecy sum rate (ESSR) of the LUs by jointly optimizing the long-term antenna positions and the short-term precoding and AN design policies. The problem is formulated as

$$(P1): \quad \max_{\mathbf{t}, \{\mathbf{w}_m(\cdot)\}, \mathbf{\Sigma}(\cdot)} \sum_{m=1}^M R_m^{\text{sec}} \quad (9a)$$

$$\text{s.t.} \quad \sum_{m=1}^M \|\mathbf{w}_m(\mathbf{H})\|^2 + \text{tr}(\mathbf{\Sigma}(\mathbf{H})) \leq P_{\text{tot}},$$

$$\mathbf{\Sigma}(\mathbf{H}) \succeq \mathbf{0}, \quad \forall \mathbf{H}, \quad (9b)$$

$$\mathbf{t}_n \in \mathcal{C}, \quad \forall n \in \mathcal{N}, \quad (9c)$$

$$\|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min}, \quad \forall n \neq i. \quad (9d)$$

Problem (P1) is challenging to solve for the following reasons. First, the short-timescale variables are policies, i.e., mappings

from the instantaneous CSI of the LUs to the precoders and the AN covariance matrix. Since the instantaneous CSI of Eve is unavailable and $\bar{R}_{e,m}$ is averaged over all CIs, the optimal policies cannot be obtained by maximizing an instantaneous secrecy rate in each CI. Second, the ergodic rates in (8) have no closed-form expressions, even for a given $\mathbf{t}$ and given policies. Third, the long-term variable $\mathbf{t}$ is coupled with the short-term policies in (9a), and constraint (9d) is non-convex. Although stochastic optimization methods based on channel samples [18] can be applied to (P1), their complexity increases with the number of samples, and the obtained solutions provide little insight into the role of the MAs. To strike a balance between performance and complexity, we restrict the short-timescale design to ZF precoding and null-space AN with a long-term power allocation in the next section, which leads to tractable expressions of the ergodic rates.

III. TWO-TIMESCALE ZF-AN DESIGN AND ERGODIC SECRECY RATE ANALYSIS

In this section, we first specify the short-timescale ZF precoding and null-space AN, based on which (P1) is transformed into a problem of the antenna positions and the long-term power allocation. Then, we derive closed-form approximations of the ergodic rates of the LUs and Eve, and reveal how the antenna positions affect the ESSR.

A. Short-Timescale ZF Precoding and Null-Space AN

Given $\mathbf{t}$ and the instantaneous CSI $\mathbf{H} = \mathbf{H}(\mathbf{t})$ in a CI, the BS adopts ZF precoding with a long-term power allocation, i.e., $\mathbf{w}_m = \sqrt{p_m} \bar{\mathbf{w}}_m$, where $p_m \geq 0$ is the transmit power allocated to LU m , and

$$\bar{\mathbf{w}}_m \triangleq \frac{\mathbf{H}(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{e}_m}{\|\mathbf{H}(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{e}_m\|} \quad (10)$$

is the normalized ZF precoding direction of LU m . Since $\mathbf{H}^H \mathbf{H}(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{e}_m = \mathbf{e}_m$, we have $\mathbf{h}_i^H \bar{\mathbf{w}}_m = 0, \forall i \neq m$, i.e., the inter-user interference is completely eliminated. Meanwhile, the AN is isotropically distributed in the null space of $\mathbf{H}^H$ [5], [20], i.e.,

$$\mathbf{\Sigma} = q \mathbf{P}_{\mathbf{H}}^{\perp}, \quad \mathbf{P}_{\mathbf{H}}^{\perp} \triangleq \mathbf{I}_N - \mathbf{H}(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H, \quad (11)$$

where $q \geq 0$ denotes the AN power per dimension. Since $\mathbf{P}_{\mathbf{H}}^{\perp}$ is the orthogonal projection matrix onto the null space of $\mathbf{H}^H$, whose dimension is $N - M$ with probability one, we have $\mathbf{H}^H \mathbf{P}_{\mathbf{H}}^{\perp} = \mathbf{0}$ and $\text{tr}(\mathbf{P}_{\mathbf{H}}^{\perp}) = N - M$. Hence, the AN causes no interference to the LUs, and the transmit power in each CI is $\sum_{m=1}^M p_m + (N - M)q$. Let $\mathbf{p} \triangleq [p_1, \dots, p_M]^T$. The power allocation $(\mathbf{p}, q)$ is optimized based on the statistical CSI and remains fixed within a frame, whereas $\{\bar{\mathbf{w}}_m\}$ and $\mathbf{P}_{\mathbf{H}}^{\perp}$ are updated in each CI based on $\mathbf{H}$.

With the above design, the SINRs in (5) and (6) reduce to

$$\gamma_m = \frac{p_m |\mathbf{h}_m^H \bar{\mathbf{w}}_m|^2}{\sigma_m^2} = \frac{p_m}{\sigma_m^2 [(\mathbf{H}^H \mathbf{H})^{-1}]_{m,m}}, \quad (12)$$

$$\gamma_{e,m} = \frac{p_m |\mathbf{g}^H \bar{\mathbf{w}}_m|^2}{q \mathbf{g}^H \mathbf{P}_{\mathbf{H}}^{\perp} \mathbf{g} + \sigma_e^2}, \quad (13)$$

respectively, where the second equality in (12) holds since $\mathbf{h}_m^H \mathbf{H} (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{e}_m = 1$ and $\|\mathbf{H} (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{e}_m\|^2 = [(\mathbf{H}^H \mathbf{H})^{-1}]_{m,m}$.

The adopted ZF-AN design has several appealing properties in the considered two-timescale framework. First, it requires only the instantaneous CSI of the LUs, which is consistent with the protocol in Section II-B. Second, it eliminates both the inter-user interference and the AN at the LUs, so that γ_m in (12) depends only on the channels of the LUs, whereas the information leakage $|\mathbf{g}^H \tilde{\mathbf{w}}_m|^2$ and the AN power $\mathbf{g}^H \mathbf{P}_{\mathbf{H}}^\perp \mathbf{g}$ received by Eve in (13) are determined by the ZF precoding directions and the AN subspace, respectively. Third, since the instantaneous CSI of Eve is unknown, the BS has no preference among the directions in the null space of $\mathbf{H}^H$, and thus distributes the AN isotropically in this subspace [4], [5]. Fourth, the power allocation $(\mathbf{p}, q)$ is adapted only to the statistical CSI, which avoids solving an optimization problem in each CI and renders the ergodic rates analytically tractable. Note that both the equal power allocation adopted in [16] and the information-AN power splitting with equal power among the LUs are special cases of the considered long-term power allocation.

With the ZF-AN design, (P1) is transformed into the following problem of $\mathbf{t}$, $\mathbf{p}$, and q :

$$(P2) : \max_{\mathbf{t}, \mathbf{p}, q} \sum_{m=1}^M R_m^{\text{sec}}(\mathbf{t}, p_m, q) \quad (14a)$$

$$\text{s.t.} \quad \sum_{m=1}^M p_m + (N - M)q \leq P_{\text{tot}}, \quad (14b)$$

$$p_m \geq 0, \quad \forall m \in \mathcal{M}, \quad q \geq 0, \quad (14c)$$

(9c), (9d),

where $R_m^{\text{sec}}(\mathbf{t}, p_m, q)$ denotes the ergodic secrecy rate in (7) with the SINRs in (12) and (13). Since the ergodic rates in (14a) still have no closed-form expressions, we derive their approximations in the sequel.

B. Ergodic Rate of the LUs

The ergodic rate of ZF precoding over Rician fading channels has been well studied in the literature [16], [17], [19]. To present the result in a form that is convenient for the subsequent analysis, we define $\mathbf{B} \triangleq \text{diag}(\frac{\beta_1}{\kappa_1+1}, \dots, \frac{\beta_M}{\kappa_M+1})$ and $\mathbf{L}(\mathbf{t}) \triangleq \tilde{\mathbf{H}}(\mathbf{t}) \mathbf{K}^{1/2}$. From (2), the channel matrix of the LUs can be written as

$$\mathbf{H}(\mathbf{t}) = (\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}) \mathbf{B}^{1/2}, \quad (15)$$

where $\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}$ is the normalized channel matrix, whose LoS component $\mathbf{L}(\mathbf{t})$ is weighted by the square roots of the Rician factors. Accordingly, we define

$$\Phi(\mathbf{t}) \triangleq \mathbf{L}^H(\mathbf{t}) \mathbf{L}(\mathbf{t}) + N \mathbf{I}_M, \quad \Omega(\mathbf{t}) \triangleq \Phi^{-1}(\mathbf{t}), \quad (16)$$

where $\Phi(\mathbf{t}) = \mathbb{E}\{(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})^H (\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})\}$ is the correlation matrix of the normalized channels of the LUs.

Lemma 1: For given $\mathbf{t}$ and p_m , the ergodic rate of LU m is approximated by [16], [19]

$$\bar{R}_m \approx \tilde{R}_m(\mathbf{t}, p_m) \triangleq \log_2(1 + \eta_m(\mathbf{t}) p_m), \quad (17)$$

where $\eta_m(\mathbf{t}) \triangleq \zeta_m(N - M)/(N\omega_m(\mathbf{t}))$, $\zeta_m \triangleq \beta_m/((\kappa_m + 1)\sigma_m^2)$, and $\omega_m(\mathbf{t}) \triangleq [\Omega(\mathbf{t})]_{m,m}$.

Proof: By applying Jensen's inequality to the convex function $\log_2(1 + 1/x)$ of $x > 0$, we have $\bar{R}_m \geq \log_2(1 + p_m/(\sigma_m^2 \mathbb{E}\{[(\mathbf{H}^H \mathbf{H})^{-1}]_{m,m}\}))$. Following [19], the noncentral Wishart matrix $(\mathbf{L} + \tilde{\mathbf{H}})^H (\mathbf{L} + \tilde{\mathbf{H}})$ is approximated by a central Wishart matrix with N degrees of freedom and the same mean $\Phi(\mathbf{t})$, whose inverse has the mean $N\Omega(\mathbf{t})/(N - M)$. Together with (15), this yields $\mathbb{E}\{[(\mathbf{H}^H \mathbf{H})^{-1}]_{m,m}\} \approx N(\kappa_m + 1)\omega_m(\mathbf{t})/(\beta_m(N - M))$, which leads to (17). ■

Lemma 1 is equivalent to the approximation in [16, Sec. IV-A], since the normalized channel correlation matrix used therein equals $\frac{1}{N}(\mathbf{K} + \mathbf{I}_M)^{-1/2} \Phi(\mathbf{t}) (\mathbf{K} + \mathbf{I}_M)^{-1/2}$. Since $\Phi(\mathbf{t}) \succeq N \mathbf{I}_M$ and $[\Omega(\mathbf{t})]_{m,m} \geq 1/[\Phi(\mathbf{t})]_{m,m} = 1/(N(\kappa_m + 1))$ [23], we have

$$\zeta_m(N - M) \leq \eta_m(\mathbf{t}) \leq \zeta_m(N - M)(\kappa_m + 1), \quad (18)$$

where the lower bound is achieved for $\kappa_m = 0$, and the upper bound is achieved if the LoS component of LU m is orthogonal to those of the other LUs, i.e., $\tilde{\mathbf{h}}_m^H(\mathbf{t}) \tilde{\mathbf{h}}_i(\mathbf{t}) = 0$ for all $i \neq m$ with $\kappa_i > 0$. Therefore, $1/(N\omega_m(\mathbf{t})) \in [1, \kappa_m + 1]$ can be interpreted as the LoS gain of LU m that survives the ZF precoding, which can be improved by decorrelating the LoS components of the LUs via antenna position optimization.

Remark 1 (Accuracy of Lemma 1): Lemma 1 is a lower-bound-type approximation. For $\kappa_m = 0$, it reduces to the well-known lower bound $\log_2(1 + \zeta_m p_m(N - M))$ for Rayleigh fading [22]. For large Rician factors, the degrees-of-freedom loss $(N - M)/N$ of the central Wishart approximation also applies to the deterministic LoS component, and thus the gap between (17) and $\bar{R}_m$ increases with κ_m and approaches $\log_2(N/(N - M))$ at high signal-to-noise ratio (SNR). Nevertheless, this gap mainly depends on the Rician factors and the ratio $N/(N - M)$, and only weakly on the antenna positions, as will be verified in Section V. Hence, it has a marginal impact on the optimization of $\mathbf{t}$, while it renders the approximate ESSR conservative. Therefore, we adopt (17) due to its simplicity and wide acceptance, and focus on the ergodic wiretap rate of Eve, for which no existing result is available.

C. Ergodic Wiretap Rate of Eve

Different from γ_m , the SINR of Eve in (13) is the ratio of the information leakage $|\mathbf{g}^H \tilde{\mathbf{w}}_m|^2$ to the AN-plus-noise power, both of which depend on the random ZF precoding direction and AN subspace induced by $\mathbf{H}$. The following lemma characterizes their statistics conditioned on $\mathbf{H}$.

Lemma 2: Conditioned on $\mathbf{H}$, the information leakage $X_m \triangleq |\mathbf{g}^H \tilde{\mathbf{w}}_m|^2$ and the AN power $Y \triangleq \mathbf{g}^H \mathbf{P}_{\mathbf{H}}^\perp \mathbf{g}$ received by Eve are independent, and their conditional means are given by

$$\mathbb{E}\{X_m | \mathbf{H}\} = \tilde{\beta}_e(\kappa_e |\tilde{\mathbf{g}}^H \tilde{\mathbf{w}}_m|^2 + 1), \quad (19)$$

$$\mathbb{E}\{Y | \mathbf{H}\} = \tilde{\beta}_e(\kappa_e \tilde{\mathbf{g}}^H \mathbf{P}_{\mathbf{H}}^\perp \tilde{\mathbf{g}} + N - M), \quad (20)$$

where $\tilde{\beta}_e \triangleq \beta_e/(\kappa_e + 1)$.

Proof: Let $\mathbf{P}_{\mathbf{H}} \triangleq \mathbf{I}_N - \mathbf{P}_{\mathbf{H}}^\perp$ denote the orthogonal projection matrix onto the column space of $\mathbf{H}$. Since $\tilde{\mathbf{w}}_m$ lies in the column space of $\mathbf{H}$, we have $\tilde{\mathbf{w}}_m = \mathbf{P}_{\mathbf{H}} \tilde{\mathbf{w}}_m$ and thus

$X_m = |(\mathbf{P}_{\mathbf{H}\mathbf{g}})^H \bar{\mathbf{w}}_m|^2$, whereas $Y = \|\mathbf{P}_{\mathbf{H}\mathbf{g}}^\perp\|^2$. Since $\tilde{\mathbf{g}}$ is independent of $\mathbf{H}$, $\mathbf{P}_{\mathbf{H}\mathbf{g}}$ and $\mathbf{P}_{\mathbf{H}\mathbf{g}}^\perp$ are jointly Gaussian given $\mathbf{H}$, with the cross-covariance matrix $\tilde{\beta}_e \mathbf{P}_{\mathbf{H}\mathbf{g}} \mathbf{P}_{\mathbf{H}\mathbf{g}}^\perp = \mathbf{0}$. Hence, $\mathbf{P}_{\mathbf{H}\mathbf{g}}$ and $\mathbf{P}_{\mathbf{H}\mathbf{g}}^\perp$ are conditionally independent, and so are X_m and Y . Moreover, (19) and (20) follow from (3), $\|\bar{\mathbf{w}}_m\| = 1$, and $\text{tr}(\mathbf{P}_{\mathbf{H}}^\perp) = N - M$. ■

Lemma 2 reveals that, for given channels of the LUs, the information leakage is determined only by the component of the wiretap channel in the signal subspace spanned by the channels of the LUs, whereas the AN received by Eve is determined only by the component in the orthogonal complement of this subspace. Moreover, conditioned on $\mathbf{H}$, Y is the sum of the energies of the projections of $\mathbf{g}$ onto $N - M$ orthonormal basis vectors of the null space of $\mathbf{H}^H$, which are independent random variables. Based on these properties, we apply to $\bar{R}_{e,m}$ the approximation $\mathbb{E}\{\log_2(1 + X/Y)\} \approx \log_2(1 + \mathbb{E}\{X\}/\mathbb{E}\{Y\})$ for nonnegative random variables X and Y [19, Lemma 1], which becomes increasingly accurate as X and Y consist of more independent nonnegative terms. By using Lemma 2 and taking the expectation over $\tilde{\mathbf{H}}$, we obtain

$$\bar{R}_{e,m} \approx \log_2 \left(1 + \frac{p_m (\kappa_e \mathbb{E}\{\|\tilde{\mathbf{g}}^H \bar{\mathbf{w}}_m\|^2\} + 1)}{q (\kappa_e \mathbb{E}\{\tilde{\mathbf{g}}^H \mathbf{P}_{\mathbf{H}\mathbf{g}}^\perp \tilde{\mathbf{g}}\} + N - M) + \bar{\sigma}_e^2} \right), \quad (21)$$

where $\bar{\sigma}_e^2 \triangleq \sigma_e^2 / \tilde{\beta}_e = \sigma_e^2 (\kappa_e + 1) / \beta_e$ is the normalized noise power of Eve, and the remaining expectations are taken over $\tilde{\mathbf{H}}$.

The two expectations in (21) are the LoS energy of Eve along the ZF precoding direction of LU m and in the AN subspace, respectively. They depend on $\mathbf{t}$ through both the LoS and the random NLoS components of the LU channels, and have no closed-form expressions. Note that $\bar{\mathbf{w}}_m$ and $\mathbf{P}_{\mathbf{H}}^\perp$ are invariant to the scaling of the columns of $\mathbf{H}$. Therefore, from (15), they can be equivalently computed based on the normalized channel matrix $\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}$. This motivates the following lemma for a generic noncentral Gaussian matrix.

Lemma 3: Let $\mathbf{F} = \bar{\mathbf{F}} + \mathbf{W} \in \mathbb{C}^{N \times k}$ with $k < N$, where $\bar{\mathbf{F}}$ is deterministic and the entries of $\mathbf{W}$ are i.i.d. $\mathcal{CN}(0, 1)$ random variables. Let $\mathbf{b} \in \mathbb{C}^{N \times 1}$ be a deterministic vector, and define $\Phi_{\mathbf{F}} \triangleq \mathbb{E}\{\mathbf{F}^H \mathbf{F}\} = \bar{\mathbf{F}}^H \bar{\mathbf{F}} + N \mathbf{I}_k$, $\bar{\mathbf{w}}_m(\mathbf{F}) \triangleq \mathbf{F}(\mathbf{F}^H \mathbf{F})^{-1} \mathbf{e}_m / \|\mathbf{F}(\mathbf{F}^H \mathbf{F})^{-1} \mathbf{e}_m\|$, and $\mathbf{P}_{\mathbf{F}}^\perp \triangleq \mathbf{I}_N - \mathbf{F}(\mathbf{F}^H \mathbf{F})^{-1} \mathbf{F}^H$. Then, we have

$$\mathbb{E}\{\|\mathbf{b}^H \bar{\mathbf{w}}_m(\mathbf{F})\|^2\} \approx \frac{|\Phi_{\mathbf{F}}^{-1} \bar{\mathbf{F}}^H \mathbf{b}|_m|^2 + \|\mathbf{b}\|^2 |\Phi_{\mathbf{F}}^{-2}|_{m,m}}{|\Phi_{\mathbf{F}}^{-1}|_{m,m}}, \quad (22)$$

$$\mathbb{E}\{\|\mathbf{b}^H \mathbf{P}_{\mathbf{F}}^\perp \mathbf{b}\}\} \approx (\|\mathbf{b}\|^2 - \mathbf{b}^H \bar{\mathbf{F}} \Phi_{\mathbf{F}}^{-1} \bar{\mathbf{F}}^H \mathbf{b}) (1 - \text{tr}(\Phi_{\mathbf{F}}^{-1})). \quad (23)$$

Moreover, (22) and (23) are exact for $\bar{\mathbf{F}} = \mathbf{0}$, and are asymptotically exact for $\bar{\mathbf{F}} = \varrho \bar{\mathbf{F}}_0$ as $\varrho \rightarrow \infty$, where $\bar{\mathbf{F}}_0$ has full column rank. The right-hand sides of (22) and (23) lie in $[0, \|\mathbf{b}\|^2]$, as do their left-hand sides.

Proof: See Appendix A. ■

Lemma 3 is based on replacing the random matrix $(\mathbf{F}^H \mathbf{F})^{-1}$ by the inverse of its mean, which is accurate when either the LoS component dominates or N is large compared with k . It is exact in the two extreme regimes, i.e., the NLoS-dominant regime in which the ZF direction and the null space

TABLE I
KEY SYMBOLS

Symbol	Definition
$\mathbf{t}, \mathcal{C}, D_{\min}$	Antenna positions, movable region, and minimum distance
$\bar{\mathbf{h}}_m(\mathbf{t}), \bar{\mathbf{g}}(\mathbf{t})$	LoS components of the channels of LU m and Eve, (1)
$\kappa_m, \kappa_e; \beta_m, \beta_e$	Rician factors; large-scale fading coefficients
p_m, q, P_{tot}	Power of LU m , AN power per dimension, and power budget
$\zeta_m, \bar{\sigma}_e^2$	$\beta_m / ((\kappa_m + 1) \sigma_m^2)$ and $\sigma_e^2 (\kappa_e + 1) / \beta_e$
$\mathbf{L}(\mathbf{t})$	Weighted LoS matrix $\bar{\mathbf{H}}(\mathbf{t}) \mathbf{K}^{1/2}$ of the LUs
$\Phi(\mathbf{t}), \Omega(\mathbf{t})$	$\mathbf{L}^H(\mathbf{t}) \mathbf{L}(\mathbf{t}) + N \mathbf{I}_M$ and its inverse, (16)
$\mathbf{v}(\mathbf{t})$	LoS correlation vector $\mathbf{L}^H(\mathbf{t}) \bar{\mathbf{g}}(\mathbf{t})$ between Eve and the LUs
$\omega_m(\mathbf{t})$	$ \Omega(\mathbf{t}) _{m,m}$, correlation of LU m with the other LUs
$c_m(\mathbf{t}), d(\mathbf{t})$	LoS energy of Eve along the ZF direction of LU m and in the AN subspace, (24), (25)
$\eta_m(\mathbf{t})$	Effective ZF gain of LU m , (17)
$\xi_m(\mathbf{t}), \psi(\mathbf{t})$	Normalized information leakage and AN power at Eve, (26)

are isotropically distributed, and the LoS-dominant regime in which they are determined by $\bar{\mathbf{F}}$. By applying Lemma 3 to (21), we obtain the following closed-form approximation of the ergodic wiretap rate.

Proposition 1: Define $\mathbf{v}(\mathbf{t}) \triangleq \mathbf{L}^H(\mathbf{t}) \bar{\mathbf{g}}(\mathbf{t}) = \mathbf{K}^{1/2} \bar{\mathbf{H}}^H(\mathbf{t}) \bar{\mathbf{g}}(\mathbf{t})$ and

$$c_m(\mathbf{t}) \triangleq \frac{|\Omega(\mathbf{t}) \mathbf{v}(\mathbf{t})|_m|^2 + N |\Omega^2(\mathbf{t})|_{m,m}}{\omega_m(\mathbf{t})}, \quad (24)$$

$$d(\mathbf{t}) \triangleq (N - \mathbf{v}^H(\mathbf{t}) \Omega(\mathbf{t}) \mathbf{v}(\mathbf{t})) (1 - \text{tr}(\Omega(\mathbf{t}))). \quad (25)$$

Then, the ergodic wiretap rate of Eve for s_m is approximated by

$$\bar{R}_{e,m} \approx \tilde{R}_{e,m}(\mathbf{t}, p_m, q) \triangleq \log_2 \left(1 + \frac{\xi_m(\mathbf{t}) p_m}{q \psi(\mathbf{t}) + \bar{\sigma}_e^2} \right), \quad (26)$$

where $\xi_m(\mathbf{t}) \triangleq \kappa_e c_m(\mathbf{t}) + 1$ and $\psi(\mathbf{t}) \triangleq \kappa_e d(\mathbf{t}) + N - M$.

Proof: From (15), $\bar{\mathbf{w}}_m = \bar{\mathbf{w}}_m(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})$ and $\mathbf{P}_{\mathbf{H}}^\perp = \mathbf{P}_{\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}}^\perp$. Applying Lemma 3 with $\bar{\mathbf{F}} = \mathbf{L}(\mathbf{t})$, $\mathbf{W} = \tilde{\mathbf{H}}$, $k = M$, and $\mathbf{b} = \bar{\mathbf{g}}(\mathbf{t})$, for which $\Phi_{\mathbf{F}} = \Phi(\mathbf{t})$, $\bar{\mathbf{F}}^H \mathbf{b} = \mathbf{v}(\mathbf{t})$, and $\|\mathbf{b}\|^2 = N$, yields $\mathbb{E}\{\|\bar{\mathbf{g}}^H \bar{\mathbf{w}}_m\|^2\} \approx c_m(\mathbf{t})$ and $\mathbb{E}\{\tilde{\mathbf{g}}^H \mathbf{P}_{\mathbf{H}}^\perp \tilde{\mathbf{g}}\} \approx d(\mathbf{t})$. Substituting them into (21) completes the proof. ■

In Proposition 1, $c_m(\mathbf{t}) \in [0, N]$ and $d(\mathbf{t}) \in [0, N]$ approximate the LoS energy of Eve along the ZF precoding direction of LU m and in the AN subspace, respectively. Accordingly, $\xi_m(\mathbf{t})$ and $\psi(\mathbf{t})$ can be interpreted as the normalized information leakage for LU m and the normalized AN power received by Eve, respectively. Different from the existing ergodic-rate approximations for ZF precoding, which only involve the diagonal entries of $\Omega(\mathbf{t})$, Proposition 1 explicitly captures the correlation between the LoS components of Eve and the LUs via $\mathbf{v}(\mathbf{t})$.

D. Approximate ESSR and Discussions

For ease of reference, the key symbols are summarized in Table I. Based on Lemma 1 and Proposition 1, the ergodic

secrecy rate of LU m is approximated by

$$R_m^{\text{sec}} \approx \tilde{R}_m^{\text{sec}}(\mathbf{t}, p_m, q) \triangleq [\tilde{R}_m(\mathbf{t}, p_m) - \tilde{R}_{e,m}(\mathbf{t}, p_m, q)]^+, \quad (27)$$

which depends only on the statistical CSI of the LUs and Eve, and is thus suitable for the large-timescale design. In the following, we discuss the insights revealed by (27).

Remark 2 (Role of the antenna positions): From (17), (24), and (25), the antenna positions affect the approximate ESSR only through three geometric quantities, i.e., $\omega_m(\mathbf{t})$, $c_m(\mathbf{t})$, and $d(\mathbf{t})$, which in turn depend on $\mathbf{t}$ only via $\Phi(\mathbf{t})$ and $\mathbf{v}(\mathbf{t})$, i.e., via the LoS array correlations

$$\begin{aligned} [\bar{\mathbf{H}}^H(\mathbf{t})\bar{\mathbf{H}}(\mathbf{t})]_{i,j} &= \sum_{n=1}^N e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T(\mathbf{a}_j - \mathbf{a}_i)}, \\ [\bar{\mathbf{H}}^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})]_i &= \sum_{n=1}^N e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T(\mathbf{a}_e - \mathbf{a}_i)}. \end{aligned} \quad (28)$$

Specifically, a smaller $\omega_m(\mathbf{t})$ indicates that the LoS component of LU m is less correlated with those of the other LUs, which yields a larger ZF gain; a smaller $c_m(\mathbf{t})$ indicates less information leakage to Eve; and a larger $d(\mathbf{t})$ indicates more AN received by Eve. In particular, if $\mathbf{v}(\mathbf{t}) = \mathbf{0}$, i.e., the LoS component of Eve is orthogonal to those of all the LUs, then $c_m(\mathbf{t})$ reduces to its minimum $N[\Omega^2(\mathbf{t})]_{m,m}/\omega_m(\mathbf{t})$ caused by the NLoS components, and $d(\mathbf{t})$ achieves its maximum $N(1 - \text{tr}(\Omega(\mathbf{t})))$ for a given $\Phi(\mathbf{t})$. Hence, by decorrelating the LoS component of Eve from those of the LUs, the MAs can simultaneously reduce the information leakage and increase the AN received by Eve, which is difficult for an FPA array when Eve is located in a direction close to that of an LU.

Remark 3 (Special cases): 1) If $\kappa_e = 0$, i.e., the channel of Eve has no LoS component, then $\tilde{R}_{e,m} = \log_2(1 + p_m/(q(N - M) + \bar{\sigma}_e^2))$ does not depend on $\mathbf{t}$. In this case, the statistical CSI of Eve carries no spatial information, and the MAs enhance the secrecy performance only by improving $\omega_m(\mathbf{t})$ as in the rate-oriented design. 2) If $M = 1$, then $\Phi(\mathbf{t}) = N(\kappa_1 + 1)$ is a scalar, and thus $\eta_1 = \zeta_1(N - 1)(\kappa_1 + 1)$ does not depend on $\mathbf{t}$.² In contrast,

$$c_1(\mathbf{t}) = \frac{\kappa_1 |\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2 / N + 1}{\kappa_1 + 1}, \quad (29)$$

$$d(\mathbf{t}) = \left(N - \frac{\kappa_1 |\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2}{N(\kappa_1 + 1)} \right) \left(1 - \frac{1}{N(\kappa_1 + 1)} \right) \quad (30)$$

depend on $\mathbf{t}$ through the LoS correlation $|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ between the LU and Eve. Hence, different from the rate-oriented two-timescale design, where the MAs bring no gain for a single user [16], the MAs enhance the secrecy performance of a single LU by reducing the LoS correlation between the LU and Eve.

Remark 4 (High-SNR behavior and AN saving): Consider the information-AN power splitting with equal power among

²In fact, for $M = 1$, the ZF precoder reduces to $\bar{\mathbf{w}}_1 = \mathbf{h}_1 / \|\mathbf{h}_1\|$, and the exact ergodic rate $\bar{R}_1 = \mathbb{E}\{\log_2(1 + p_1 \|\mathbf{h}_1\|^2 / \sigma_1^2)\}$ does not depend on $\mathbf{t}$, since the distribution of $\|\mathbf{h}_1\|^2$ depends on $\mathbf{h}_1(\mathbf{t})$ only through $\|\mathbf{h}_1(\mathbf{t})\|^2 = N$.

the LUs, i.e., $p_m = \alpha P_{\text{tot}}/M$ and $q = (1 - \alpha)P_{\text{tot}}/(N - M)$ with $\alpha \in (0, 1)$. As $P_{\text{tot}} \rightarrow \infty$, we have

$$\tilde{R}_m - \tilde{R}_{e,m} = \log_2 \frac{\alpha P_{\text{tot}} \eta_m(\mathbf{t})}{M} - \log_2 \left(1 + \frac{\alpha \Theta_m(\mathbf{t})}{1 - \alpha} \right) + o(1), \quad (31)$$

where $\Theta_m(\mathbf{t}) \triangleq \frac{(N-M)\xi_m(\mathbf{t})}{M\psi(\mathbf{t})}$ is the ratio of the normalized information leakage to the normalized AN power received by Eve. Hence, the high-SNR slope of the ESSR equals M regardless of $\mathbf{t}$, and the antenna positions affect the ESSR through the constant term. Moreover, the value of α that maximizes the sum of the constant terms in (31) over m is the root of $\sum_{m=1}^M \left[\frac{1}{\alpha} - \frac{1}{1-\alpha} - \frac{\Theta_m-1}{1+\alpha(\Theta_m-1)} \right] = 0$, which, for $\Theta_m = \Theta$, $\forall m$, is given by

$$\alpha^* = \frac{1}{1 + \sqrt{\Theta}}. \quad (32)$$

Since the MAs can decrease $c_m(\mathbf{t})$ and increase $d(\mathbf{t})$, they decrease Θ and thus increase α^* . In other words, the MAs reduce the AN power required for secure transmission, which is referred to as the AN-saving effect.

IV. PROPOSED TWO-TIMESCALE OPTIMIZATION ALGORITHM

In this section, based on the approximate ESSR derived in Section III, we reformulate (P2) as a deterministic problem that depends only on the statistical CSI, and develop a low-complexity AO algorithm to solve it.

A. Problem Reformulation

By replacing the ergodic secrecy rates in (P2) with their approximations in (27), we consider the following problem:

$$\begin{aligned} (\text{P3}) : \max_{\mathbf{t}, \mathbf{p}, q} \quad & F(\mathbf{t}, \mathbf{p}, q) \triangleq \sum_{m=1}^M \left[\log_2(1 + \eta_m(\mathbf{t})p_m) \right. \\ & \left. - \log_2 \left(1 + \frac{\xi_m(\mathbf{t})p_m}{q\psi(\mathbf{t}) + \bar{\sigma}_e^2} \right) \right] \quad (33a) \\ \text{s.t.} \quad & (9c), (9d), (14b), (14c), \end{aligned}$$

where the operator $[\cdot]^+$ in (27) is dropped without loss of optimality, as shown in the following proposition.

Proposition 2: The optimal value of (P3) equals that of the problem obtained by replacing $F(\mathbf{t}, \mathbf{p}, q)$ in (33a) with $\sum_{m=1}^M \tilde{R}_m^{\text{sec}}(\mathbf{t}, p_m, q)$. Moreover, at any optimal solution of (P3), every term in the summation of (33a) is nonnegative.

Proof: Since $[x]^+ \geq x$, the optimal value with $[\cdot]^+$ is no smaller than that of (P3). Conversely, for any feasible $(\mathbf{t}, \mathbf{p}, q)$, let $\mathbf{p}'$ be obtained by setting $p'_m = 0$ for all m with $\tilde{R}_m(\mathbf{t}, p_m) < \tilde{R}_{e,m}(\mathbf{t}, p_m, q)$ and $p'_m = p_m$ otherwise. Then, $(\mathbf{t}, \mathbf{p}', q)$ is feasible, and since the m -th term of F depends only on p_m and q for a given $\mathbf{t}$, we have $F(\mathbf{t}, \mathbf{p}', q) = \sum_{m=1}^M \tilde{R}_m^{\text{sec}}(\mathbf{t}, p_m, q)$. Hence, the two optimal values are equal. If some term were negative at an optimal solution of (P3), setting the corresponding p_m to zero would strictly increase F , which is a contradiction. ■

Proposition 2 shows that the non-smooth operator $[\cdot]^+$ can be removed by allowing the BS to switch off the LUs with non-positive secrecy rates, which is enabled by the per-user

power allocation $\mathbf{p}$. In contrast, under equal power allocation among the LUs, the power allocated to such LUs is wasted. Problem (P3) is still non-convex due to the coupling between $\mathbf{t}$ and $(\mathbf{p}, q)$ in (33a) and the non-convex constraint (9d). To tackle these difficulties, we adopt the AO framework, which alternately optimizes $(\mathbf{p}, q)$ for given $\mathbf{t}$ and $\mathbf{t}$ for given $(\mathbf{p}, q)$.

B. Power Allocation Optimization

For given $\mathbf{t}$, the coefficients η_m , ξ_m , and ψ are constants, and (P3) reduces to

$$(P3.1) : \max_{\mathbf{p}, q} \sum_{m=1}^M f_m(p_m; q) \quad \text{s.t. (14b), (14c),} \quad (34)$$

where

$$f_m(p; q) \triangleq \log_2(1 + \eta_m p) - \log_2(1 + \vartheta_m(q) p), \quad (35)$$

and $\vartheta_m(q) \triangleq \xi_m / (q\psi + \bar{\sigma}_e^2)$ is the effective channel gain of Eve for LU m , which decreases with q . We first optimize $\mathbf{p}$ for a given q . The first-order and second-order derivatives of f_m with respect to (w.r.t.) p are given by

$$\frac{\partial f_m}{\partial p} = \frac{\eta_m - \vartheta_m}{\ln 2 (1 + \eta_m p)(1 + \vartheta_m p)}, \quad (36)$$

$$\frac{\partial^2 f_m}{\partial p^2} = \frac{1}{\ln 2} \left[\frac{\vartheta_m^2}{(1 + \vartheta_m p)^2} - \frac{\eta_m^2}{(1 + \eta_m p)^2} \right], \quad (37)$$

where the argument q of ϑ_m is omitted for brevity. If $\eta_m \leq \vartheta_m$, then $f_m(p; q) \leq 0$ for all $p \geq 0$, and thus $p_m = 0$ is optimal. If $\eta_m > \vartheta_m$, then $\eta_m(1 + \vartheta_m p) - \vartheta_m(1 + \eta_m p) = \eta_m - \vartheta_m > 0$, i.e., $\frac{\eta_m}{1 + \eta_m p} > \frac{\vartheta_m}{1 + \vartheta_m p}$, which implies that f_m is strictly increasing and strictly concave in $p \geq 0$. Let $\mathcal{M}_a(q) \triangleq \{m \in \mathcal{M} \mid \eta_m > \vartheta_m(q)\}$ denote the set of LUs that can achieve positive secrecy rates, and let $P_1(q) \triangleq P_{\text{tot}} - (N - M)q$ denote the power available for the confidential signals. Then, for a given q , the optimization of $\mathbf{p}$ is a convex problem, whose solution is given by the following proposition.

Proposition 3 (Secrecy water-filling): For a given q with $P_1(q) > 0$ and $\mathcal{M}_a(q) \neq \emptyset$, the optimal power allocation is $p_m^* = 0$ for $m \notin \mathcal{M}_a(q)$ and

$$p_m^*(\nu) = \left[\frac{2(\rho_m(\nu) - 1)}{\sqrt{(\eta_m - \vartheta_m)^2 + 4\eta_m\vartheta_m\rho_m(\nu)} + \eta_m + \vartheta_m} \right]^+ \quad (38)$$

for $m \in \mathcal{M}_a(q)$, where $\rho_m(\nu) \triangleq (\eta_m - \vartheta_m)/(\nu \ln 2)$, and the water level $\nu > 0$ is the unique solution of $\sum_{m \in \mathcal{M}_a(q)} p_m^*(\nu) = P_1(q)$.

Proof: Since the objective is strictly concave and increasing in $\{p_m\}_{m \in \mathcal{M}_a(q)}$ and Slater's condition holds, the Karush–Kuhn–Tucker (KKT) conditions are necessary and sufficient for optimality, and the power constraint is active at the optimum [24]. Let $\nu > 0$ denote the dual variable associated with the power constraint. The KKT conditions yield $\partial f_m / \partial p = \nu$ if $p_m^* > 0$ and $\partial f_m / \partial p|_{p=0} \leq \nu$ otherwise. From (36), the former is equivalent to $(1 + \eta_m p)(1 + \vartheta_m p) = \rho_m(\nu)$, i.e., a quadratic equation in p , whose positive root, written in a numerically stable form, is given by (38). The latter is equivalent to $\rho_m(\nu) \leq 1$, in which case (38) gives $p_m^* = 0$. Finally, since $\sum_{m \in \mathcal{M}_a(q)} p_m^*(\nu)$ is continuous and

Algorithm 1 Power Allocation for Given $\mathbf{t}$

Require: $\{\eta_m, \xi_m\}_{m \in \mathcal{M}}$, ψ , P_{tot} , and grid size Q .

- 1: **for** each q in a uniform grid of Q points over $[0, q_{\max}]$ **do**
- 2: Compute $\vartheta_m(q)$ and $\mathcal{M}_a(q)$.
- 3: Find ν by bisection such that $\sum_{m \in \mathcal{M}_a(q)} p_m^*(\nu) = P_1(q)$, where $p_m^*(\nu)$ is given by (38).
- 4: Evaluate $G(q)$.
- 5: **end for**
- 6: Repeat Steps 1–5 over a uniform grid of Q points between the two neighbors of the best grid point.

Ensure: q^* and the corresponding $\mathbf{p}^*$.

strictly decreasing in ν over $(0, \max_{m \in \mathcal{M}_a(q)} (\eta_m - \vartheta_m) / \ln 2)$, ranging from $+\infty$ to 0, the water level ν satisfying the power constraint with equality is unique. ■

Note that (38) reduces to the classic water-filling solution $[1/(\nu \ln 2) - 1/\eta_m]^+$ when $\vartheta_m \rightarrow 0$. Hence, (38) can be regarded as a secrecy water-filling solution, in which the LUs whose effective channel gains do not exceed that of Eve are switched off. Since $p_m^*(\nu)$ is non-increasing in ν , the water level can be efficiently found by the bisection method.

Next, let $G(q) \triangleq \sum_{m \in \mathcal{M}} f_m(p_m^*; q)$ denote the objective value achieved by the optimal power allocation for a given q , where $G(q) = 0$ if $\mathcal{M}_a(q) = \emptyset$ or $P_1(q) = 0$. The optimal AN power per dimension is obtained by solving the one-dimensional problem

$$q^* = \arg \max_{0 \leq q \leq q_{\max}} G(q), \quad (39)$$

where $q_{\max} \triangleq P_{\text{tot}} / (N - M)$. Increasing q reduces the effective channel gains $\vartheta_m(q)$ of Eve at the cost of less power for the confidential signals, and (39) finds the best tradeoff. Since $G(q)$ is not necessarily unimodal, (39) is solved by a coarse grid search over $[0, q_{\max}]$ followed by a fine grid search around the best grid point. The overall power allocation procedure is summarized in Algorithm 1.

C. Antenna Position Optimization

For given $(\mathbf{p}, q)$, (P3) reduces to

$$(P3.2) : \max_{\mathbf{t}} F(\mathbf{t}) \quad \text{s.t. (9c), (9d),} \quad (40)$$

where $F(\mathbf{t}) \triangleq F(\mathbf{t}, \mathbf{p}, q)$ for brevity. Problem (P3.2) is non-convex since $F(\mathbf{t})$ is neither concave nor convex w.r.t. $\mathbf{t}$ and constraint (9d) is non-convex. Similar to [16], we optimize the position of each MA in an element-wise manner with the positions of the other MAs being fixed, and update $\mathbf{t}_n$ by the MM method [26].

1) *Dependence on $\mathbf{t}_n$:* Let $\mathbf{l}_n(\mathbf{t}_n) \in \mathbb{C}^{M \times 1}$ denote the conjugate transpose of the n -th row of $\mathbf{L}(\mathbf{t})$, i.e.,

$$\mathbf{l}_n(\mathbf{t}_n) = \mathbf{K}^{1/2} \left[e^{-j\frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_1}, \dots, e^{-j\frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_M} \right]^T. \quad (41)$$

Since $\mathbf{L}^H \mathbf{L} = \sum_{i=1}^N \mathbf{l}_i \mathbf{l}_i^H$ and $\mathbf{L}^H \mathbf{g} = \sum_{i=1}^N \mathbf{l}_i e^{j\frac{2\pi}{\lambda} \mathbf{t}_i^T \mathbf{a}_e}$, $\Phi(\mathbf{t})$ and $\mathbf{v}(\mathbf{t})$ can be expressed as

$$\Phi(\mathbf{t}) = \mathbf{\Gamma}_n + \mathbf{l}_n(\mathbf{t}_n) \mathbf{l}_n^H(\mathbf{t}_n), \quad (42)$$

$$\mathbf{v}(\mathbf{t}) = \mathbf{r}_n + \mathbf{l}_n(\mathbf{t}_n) e^{j\frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_e}, \quad (43)$$

where $\mathbf{\Gamma}_n \triangleq \sum_{i \neq n} \mathbf{l}_i(\mathbf{t}_i) \mathbf{l}_i^H(\mathbf{t}_i) + N \mathbf{I}_M$ and $\mathbf{r}_n \triangleq \sum_{i \neq n} \mathbf{l}_i(\mathbf{t}_i) e^{j \frac{2\pi}{\lambda} \mathbf{t}_i^T \mathbf{a}_e}$ do not depend on $\mathbf{t}_n$. By the Sherman–Morrison formula [23],

$$\mathbf{\Omega}(\mathbf{t}) = \mathbf{\Gamma}_n^{-1} - \frac{\mathbf{\Gamma}_n^{-1} \mathbf{l}_n(\mathbf{t}_n) \mathbf{l}_n^H(\mathbf{t}_n) \mathbf{\Gamma}_n^{-1}}{1 + \mathbf{l}_n^H(\mathbf{t}_n) \mathbf{\Gamma}_n^{-1} \mathbf{l}_n(\mathbf{t}_n)}. \quad (44)$$

Hence, for given $\mathbf{\Gamma}_n^{-1}$ and $\mathbf{r}_n$, evaluating ω_m , c_m , d , and thus F as functions of $\mathbf{t}_n$ requires only $\mathcal{O}(M^2)$ operations.

2) *Gradient*: For $u \in \{x_n, y_n\}$, let $\mathbf{D}_u \triangleq \text{diag}(a_{1,u}, \dots, a_{M,u})$, where $a_{m,x} \triangleq \cos \theta_m \sin \phi_m$ and $a_{m,y} \triangleq \sin \theta_m$ are the entries of $\mathbf{a}_m$, and let $a_{e,u}$ denote the corresponding entry of $\mathbf{a}_e$. From (41)–(43), the derivatives of $\mathbf{l}_n$, $\mathbf{v}$, and $\mathbf{\Omega}$ w.r.t. u are given by

$$\dot{\mathbf{l}}_n \triangleq \frac{\partial \mathbf{l}_n}{\partial u} = -j \frac{2\pi}{\lambda} \mathbf{D}_u \mathbf{l}_n, \quad (45)$$

$$\dot{\mathbf{v}} \triangleq \frac{\partial \mathbf{v}}{\partial u} = j \frac{2\pi}{\lambda} (a_{e,u} \mathbf{I}_M - \mathbf{D}_u) \mathbf{l}_n e^{j \frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_e}, \quad (46)$$

$$\dot{\mathbf{\Omega}} \triangleq \frac{\partial \mathbf{\Omega}}{\partial u} = -\mathbf{\Omega}(\dot{\mathbf{l}}_n \mathbf{l}_n^H + \mathbf{l}_n \dot{\mathbf{l}}_n^H) \mathbf{\Omega}, \quad (47)$$

where (47) follows from $\partial \mathbf{\Phi}^{-1} = -\mathbf{\Phi}^{-1}(\partial \mathbf{\Phi}) \mathbf{\Phi}^{-1}$. Based on (45)–(47), the derivatives $\dot{\omega}_m \triangleq \partial \omega_m / \partial u$, $\dot{c}_m \triangleq \partial c_m / \partial u$, and $\dot{d} \triangleq \partial d / \partial u$ are derived in closed form in Appendix B. Let $\chi_m \triangleq \xi_m p_m$ and $\Upsilon \triangleq q\psi + \bar{\sigma}_e^2$, whose derivatives are $\dot{\chi}_m = \kappa_e p_m \dot{c}_m$ and $\dot{\Upsilon} = \kappa_e q \dot{d}$, respectively. Moreover, $\dot{\eta}_m = -\eta_m \dot{\omega}_m / \omega_m$. By the chain rule, the derivative of F w.r.t. u is given by

$$\frac{\partial F}{\partial u} = \frac{1}{\ln 2} \sum_{m=1}^M \left[\frac{p_m \dot{\eta}_m}{1 + \eta_m p_m} - \frac{\kappa_e p_m \dot{c}_m}{\chi_m + \Upsilon} + \frac{\kappa_e q \chi_m \dot{d}}{\Upsilon(\chi_m + \Upsilon)} \right], \quad (48)$$

and the gradient of F w.r.t. $\mathbf{t}_n$ is $\nabla_{\mathbf{t}_n} F = [\partial F / \partial x_n, \partial F / \partial y_n]^T$.

3) *Surrogate function*: With a slight abuse of notation, let $F(\mathbf{t}_n)$ denote F as a function of $\mathbf{t}_n$, with the positions of the other MAs being fixed at their latest values. Since $F(\mathbf{t}_n)$ is twice continuously differentiable and $\mathcal{C}$ is compact, $\nabla_{\mathbf{t}_n} F$ is Lipschitz continuous over $\mathcal{C}$ with some constant $\mathcal{L}_n$. According to the descent lemma [25, Lemma 5.7], for any $\delta_n \geq \mathcal{L}_n$, the following quadratic function is a lower bound of $F(\mathbf{t}_n)$ that is tight at the local point $\mathbf{t}_n^{(\ell)}$ in the ℓ -th iteration:

$$F(\mathbf{t}_n) \geq \hat{F}_n(\mathbf{t}_n | \mathbf{t}_n^{(\ell)}) \triangleq F(\mathbf{t}_n^{(\ell)}) + \nabla_{\mathbf{t}_n} F(\mathbf{t}_n^{(\ell)})^T (\mathbf{t}_n - \mathbf{t}_n^{(\ell)}) - \frac{\delta_n}{2} \|\mathbf{t}_n - \mathbf{t}_n^{(\ell)}\|^2. \quad (49)$$

Since $\mathcal{L}_n$ is difficult to obtain in closed form, δ_n is determined by backtracking [25], i.e., starting from an initial value δ_n^{ini} , δ_n is doubled until the solution of the subproblem below satisfies $F(\mathbf{t}_n^{(\ell+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(\ell+1)} | \mathbf{t}_n^{(\ell)})$, which is guaranteed once $\delta_n \geq \mathcal{L}_n$. To reduce the number of backtracking trials, we set $\delta_n^{\text{ini}} = \max\{\delta_0, \delta_n^{\text{pre}}/2\}$, where $\delta_0 > 0$ is a prescribed constant and δ_n^{pre} is the value of δ_n accepted in the previous iteration.

4) *Minimum-distance constraints*: Since $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ is convex w.r.t. $\mathbf{t}_n$, it is lower-bounded by its first-order Taylor expansion at $\mathbf{t}_n^{(\ell)}$ [16], i.e.,

$$\|\mathbf{t}_n - \mathbf{t}_i\|^2 \geq \|\Delta_{n,i}^{(\ell)}\|^2 + 2(\Delta_{n,i}^{(\ell)})^T (\mathbf{t}_n - \mathbf{t}_n^{(\ell)}), \quad (50)$$

Algorithm 2 Two-Timescale AO Algorithm for (P3)

Require: Statistical CSI, P_{tot} , $\mathcal{C}$, $D_{\min}$, δ_0 , and tolerance ϵ .
1: Initialize a feasible $\mathbf{t}^{(0)}$, obtain $(\mathbf{p}^{(0)}, q^{(0)})$ by Algorithm 1, and set $\ell = 0$.
2: **repeat**
3: **for** $n = 1$ to N **do**
4: Compute $\nabla_{\mathbf{t}_n} F$ by (48) and set $\delta_n = \delta_n^{\text{ini}}$.
5: **repeat**
6: Solve (P3.2.n) to obtain $\mathbf{t}_n^{(\ell+1)}$.
7: **if** $F(\mathbf{t}_n^{(\ell+1)}) < \hat{F}_n(\mathbf{t}_n^{(\ell+1)} | \mathbf{t}_n^{(\ell)})$ **then** $\delta_n \leftarrow 2\delta_n$.
8: **until** $F(\mathbf{t}_n^{(\ell+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(\ell+1)} | \mathbf{t}_n^{(\ell)})$
9: **end for**
10: Obtain $(\mathbf{p}', q')$ by Algorithm 1 for $\mathbf{t}^{(\ell+1)}$.
11: **If** $F(\mathbf{t}^{(\ell+1)}, \mathbf{p}', q') \geq F(\mathbf{t}^{(\ell+1)}, \mathbf{p}^{(\ell)}, q^{(\ell)})$,
12: **set** $(\mathbf{p}^{(\ell+1)}, q^{(\ell+1)}) = (\mathbf{p}', q')$; **otherwise**, **set** $(\mathbf{p}^{(\ell+1)}, q^{(\ell+1)}) = (\mathbf{p}^{(\ell)}, q^{(\ell)})$.
13: $\ell \leftarrow \ell + 1$.
14: **until** the fractional increase of F is below ϵ
Ensure: $\mathbf{t}^* = \mathbf{t}^{(\ell)}$, $\mathbf{p}^* = \mathbf{p}^{(\ell)}$, and $q^* = q^{(\ell)}$.

where $\Delta_{n,i}^{(\ell)} \triangleq \mathbf{t}_n^{(\ell)} - \mathbf{t}_i$. By replacing $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ in (9d) with the right-hand side of (50), we obtain the linear constraint

$$(\Delta_{n,i}^{(\ell)})^T \mathbf{t}_n \geq (\Delta_{n,i}^{(\ell)})^T \mathbf{t}_n^{(\ell)} + \frac{D_{\min}^2 - \|\Delta_{n,i}^{(\ell)}\|^2}{2}, \quad \forall i \neq n, \quad (51)$$

whose feasible set is a convex subset of that of (9d) w.r.t. $\mathbf{t}_n$.

5) *Subproblem for MA n* : Based on (49) and (51), $\mathbf{t}_n$ is updated by solving

$$(\text{P3.2.n}) : \max_{\mathbf{t}_n} \hat{F}_n(\mathbf{t}_n | \mathbf{t}_n^{(\ell)}) \quad \text{s.t. } \mathbf{t}_n \in \mathcal{C}, \quad (52)$$

By completing the square, (P3.2.n) is equivalent to projecting $\mathbf{t}_n^{(\ell)} + \nabla_{\mathbf{t}_n} F(\mathbf{t}_n^{(\ell)}) / \delta_n$ onto the convex polygon defined by $\mathcal{C}$ and (51), which is a convex quadratic program with two variables. Its optimal solution lies either at the projected point itself (if feasible), on an edge, or at a vertex of the polygon, and can thus be obtained exactly by enumerating the projections onto the $N + 3$ constraint lines and their pairwise intersections, or by standard solvers. Since $\mathbf{t}_n^{(\ell)}$ is feasible for (P3.2.n), the update satisfies

$$F(\mathbf{t}_n^{(\ell+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(\ell+1)} | \mathbf{t}_n^{(\ell)}) \geq \hat{F}_n(\mathbf{t}_n^{(\ell)} | \mathbf{t}_n^{(\ell)}) = F(\mathbf{t}_n^{(\ell)}), \quad (53)$$

and $\mathbf{t}^{(\ell+1)}$ is feasible for (P3.2) due to (50). Hence, the objective value of (P3.2) is non-decreasing after each element-wise update.

D. Overall Algorithm, Convergence, and Complexity

The overall algorithm for solving (P3) is summarized in Algorithm 2. Since (P3) is non-convex, the obtained solution depends on the initial antenna positions. To alleviate this issue, Algorithm 2 can be executed from multiple initial positions, e.g., the FPA array and several random feasible positions, and the solution with the largest objective value is selected.

Convergence: In each iteration of Algorithm 2, the element-wise position updates do not decrease the objective value of

(P3) according to (53), and the power allocation is updated only if the objective value does not decrease. Hence, the objective value of (P3) is non-decreasing over the iterations. Moreover, it is upper-bounded due to the finite transmit power, e.g., $F \leq \sum_{m=1}^M \log_2(1 + \zeta_m(N-M)(\kappa_m+1)P_{\text{tot}})$ according to (18) and $p_m \leq P_{\text{tot}}$. Therefore, Algorithm 2 is guaranteed to converge.

Complexity: In each iteration of Algorithm 2, forming Φ and Ω requires $\mathcal{O}(NM^2 + M^3)$ operations. For each MA, Γ_n^{-1} is obtained from Ω by a rank-one update with $\mathcal{O}(M^2)$ operations, and the gradient in (48) and each evaluation of F require $\mathcal{O}(M^2)$ operations based on (44). Solving (P3.2.n) by enumeration requires $\mathcal{O}(N^3)$ operations. Algorithm 1 requires $\mathcal{O}(QI_bM)$ operations, where I_b is the number of bisection iterations. Hence, the complexity of Algorithm 2 is $\mathcal{O}(I_o(NM^2 + M^3 + NJ(M^2 + N^3) + QI_bM))$, where I_o is the number of iterations and J is the average number of backtracking trials. Note that the complexity does not depend on the number of channel realizations, in contrast to the stochastic optimization methods based on channel samples [18]. Moreover, Algorithm 2 is executed only once per frame, whereas the ZF precoding and AN generation in each CI require $\mathcal{O}(NM^2 + M^3)$ operations.

V. NUMERICAL RESULTS

In this section, we provide numerical results to validate the accuracy of the derived approximation and to evaluate the performance of the proposed two-timescale design.

A. Simulation Setup and Benchmarks

Unless otherwise specified, the BS is equipped with $N = 8$ MAs to serve $M = 3$ LUs, the movable region is a square with side length $A = 4\lambda$, and $D_{\min} = \lambda/2$. The elevation and azimuth AoDs of the LUs are independently and uniformly distributed within $[-\pi/3, \pi/3]$. To model a challenging eavesdropping scenario, Eve is located in a direction close to that of LU 1, i.e., $\theta_e = \theta_1 + \Delta \cos \tilde{\phi}$ and $\phi_e = \phi_1 + \Delta \sin \tilde{\phi}$, where $\tilde{\phi}$ is uniformly distributed over $[0, 2\pi)$ and $\Delta = 0.1$ rad denotes the angular offset between Eve and LU 1. The Rician factors are $\kappa_m = \kappa_e = 10$ dB, the large-scale fading coefficients are normalized as $\beta_m = \beta_e = 1$, and the noise powers are $\sigma_m^2 = \sigma_e^2 = \sigma^2$, such that P_{tot}/σ^2 represents the transmit SNR, which is set to 20 dB. For Algorithm 2, we set $\delta_0 = 1$, $\epsilon = 10^{-4}$, $Q = 41$, and 60 bisection iterations, and select the best solution obtained from three initializations, i.e., a 2×4 uniform planar array (UPA) with $\lambda/2$ spacing centered at the origin and two random feasible positions. All results are averaged over 50 random realizations of the AoDs. For each obtained solution, the ESSR is evaluated by Monte Carlo simulations over 5×10^3 channel realizations with the exact ZF precoding and null-space AN in (10)–(13), i.e., the approximation is used only for optimization. The proposed design is compared with the following benchmarks:

- *MA, equal power:* The antenna positions are optimized by Algorithm 2, where the LUs are allocated equal power, i.e., $p_m = \alpha P_{\text{tot}}/M$ and $q = (1 - \alpha)P_{\text{tot}}/(N - M)$, and

only the power splitting factor $\alpha \in [0, 1]$ is optimized by a one-dimensional search.

- *MA without AN:* The antenna positions and $\mathbf{p}$ are optimized by Algorithm 2 with $q = 0$.
- *FPA:* The BS adopts the 2×4 UPA with $\lambda/2$ spacing, and $(\mathbf{p}, q)$ is optimized by Algorithm 1.
- *Sparse FPA:* The BS adopts a 2×4 UPA that spans the entire region $\mathcal{C}$, i.e., with spacings $A/3$ and A along the x - and y -axes, respectively, and $(\mathbf{p}, q)$ is optimized by Algorithm 1.
- *FPA without AN:* The $\lambda/2$ UPA with $\mathbf{p}$ optimized by Algorithm 1 and $q = 0$.

B. Accuracy of the Approximation

We first evaluate the accuracy of the closed-form approximations in Lemma 1 and Proposition 1 for two antenna configurations, i.e., the $\lambda/2$ UPA and random feasible positions in $\mathcal{C}$, under the power splitting $p_m = \alpha P_{\text{tot}}/M$ and $q = (1 - \alpha)P_{\text{tot}}/(N - M)$ with $\alpha = 0.5$. Figs. 3 and 4 show the ergodic sum rates of the LUs and Eve versus the transmit SNR and the Rician factor, respectively. It is observed from Figs. 3(b) and 4(b) that the closed-form ergodic wiretap rate in Proposition 1 matches the Monte Carlo results well for both configurations over the entire SNR and Rician-factor ranges, and the errors of the ergodic wiretap sum rate are within 0.2 bps/Hz. This validates Lemma 3 and the approximation in (21). Moreover, the ergodic wiretap rate of the random MA positions is lower than that of the UPA, since the random positions enlarge the array aperture and thus reduce the LoS correlation between Eve and LU 1. As expected from Remark 1, Lemma 1 lower-bounds the ergodic rates of the LUs, and the gap increases with κ . Nevertheless, for $\kappa = 10$ dB and $P_{\text{tot}}/\sigma^2 = 20$ dB, the gaps of the ergodic sum rate are 1.53 and 1.66 bps/Hz (i.e., about 0.5 bps/Hz per LU) for the two configurations, respectively, which confirms that the gap depends only weakly on the antenna positions.

C. Convergence of the Proposed Algorithm

Fig. 5 shows the objective value of (P3) versus the number of iterations of Algorithm 2 for one random realization of the AoDs, three initializations, and $P_{\text{tot}}/\sigma^2 \in \{10, 20, 30\}$ dB. It is observed that the objective value increases monotonically for all initializations, which verifies the convergence analysis in Section IV-D. A significant improvement is achieved in the first iteration, after which the objective value increases gradually as the antenna positions are fine-tuned, and it reaches within 1% of its final value within 60 iterations in all cases. Moreover, the final objective values obtained from different initializations differ by less than 5%, which indicates that a few initializations suffice in practice.

D. ESSR Versus Transmit SNR

Fig. 6 shows the ESSR versus the transmit SNR. The proposed design achieves the highest ESSR over the entire SNR range. For example, at $P_{\text{tot}}/\sigma^2 = 20$ dB, the proposed design improves the ESSR by 28.1% and 13.6% compared with the

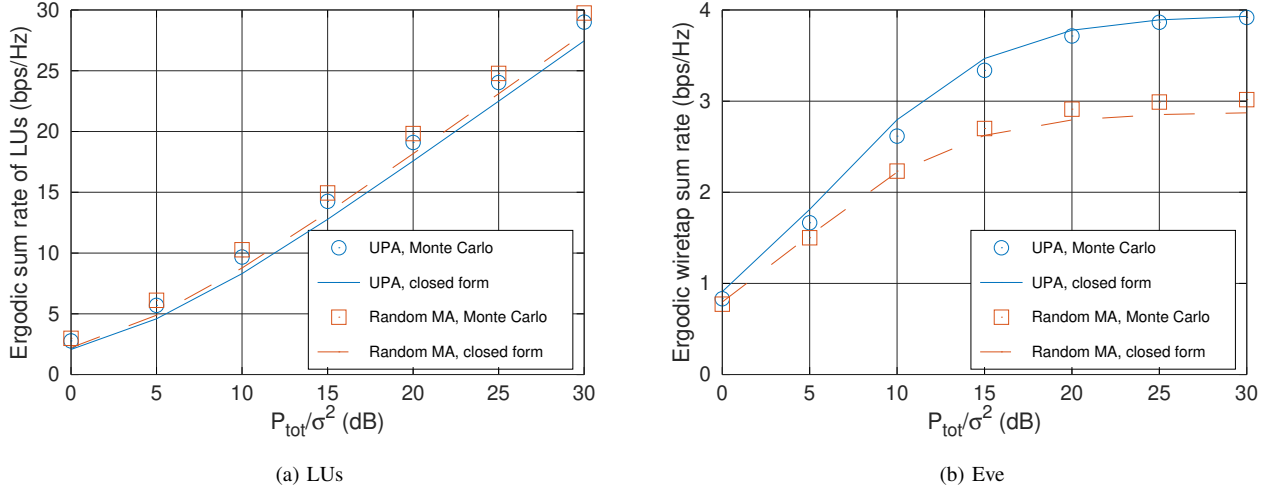


Fig. 3. Ergodic sum rates of (a) the LUs and (b) Eve versus the transmit SNR, where $\kappa_m = \kappa_e = 10$ dB and $\alpha = 0.5$.

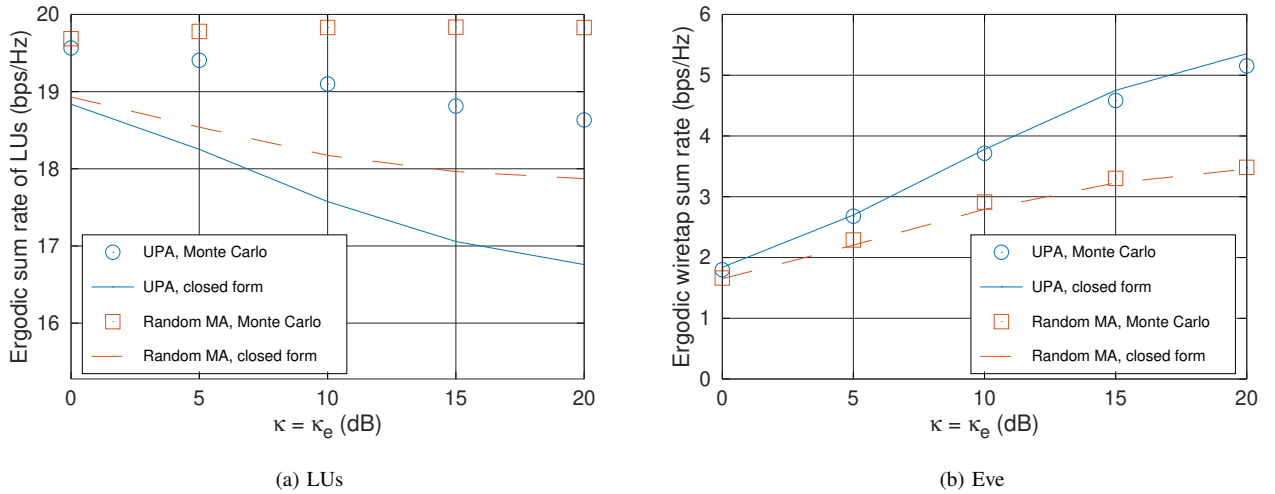


Fig. 4. Ergodic sum rates of (a) the LUs and (b) Eve versus the Rician factor $\kappa = \kappa_m = \kappa_e$, where $P_{\text{tot}}/\sigma^2 = 20$ dB and $\alpha = 0.5$.

FPA and the sparse FPA, respectively. As the SNR increases, the absolute gain of the MA over the FPA increases and then saturates at about 4.6 bps/Hz, i.e., the two curves become parallel at high SNR. This is consistent with Remark 4: the antenna positions do not change the high-SNR slope of the ESSR but improve its constant term. In contrast, the ESSRs of the schemes without AN saturate at high SNR, since the ergodic wiretap rates of Eve increase with the transmit power as fast as the ergodic rates of the LUs. Nevertheless, the MA without AN outperforms the FPA without AN by 23.4% at $P_{\text{tot}}/\sigma^2 = 20$ dB owing to the reduced information leakage. At low SNR, the proposed design allocates little or no power to the AN, and thus performs similarly to the MA without AN. Moreover, the per-user power allocation and the equal power allocation achieve similar ESSRs in this symmetric setup, which will be further discussed in Section V-F.

Fig. 7 shows the optimized AN power fraction $(N - M)q/P_{\text{tot}}$. The AN power fraction increases with the SNR and saturates at high SNR, which agrees with the asymptotically

optimal power splitting in (32). More importantly, the MA requires considerably less AN power than the FPA-based designs, e.g., 0.30 versus 0.47 for the FPA and 0.40 for the sparse FPA at $P_{\text{tot}}/\sigma^2 = 20$ dB. This verifies the AN-saving effect revealed in Remark 4: by decorrelating the LoS component of Eve from those of the LUs, the MAs reduce the information leakage and increase the AN received by Eve, which allows more power to be allocated to the confidential signals.

E. Impact of the Angular Offset and the Region Size

Fig. 8 shows the ESSR versus the angular offset Δ between Eve and LU 1. When Δ is small, e.g., $\Delta = 0.02$ rad, Eve is almost aligned with LU 1, and their LoS components can hardly be decorrelated by any array whose aperture is limited by the region size. Hence, all the schemes achieve relatively low ESSRs, and the gain of the MA over the FPA is 10.7%. As Δ increases, the ESSR of the proposed design increases rapidly and saturates for $\Delta \geq 0.3$ rad, whereas that of the

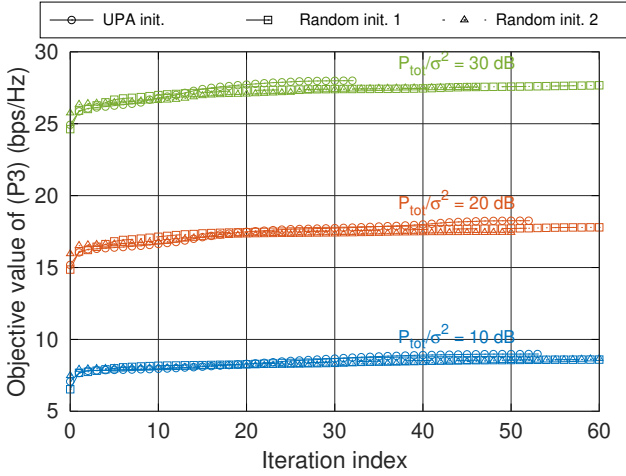


Fig. 5. Convergence of Algorithm 2 for different initializations and transmit SNRs.

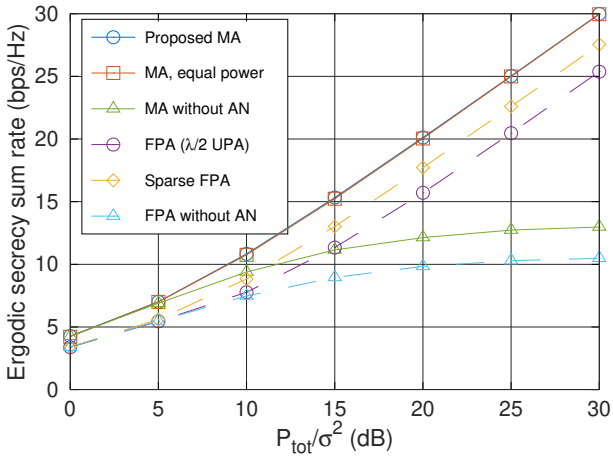


Fig. 6. ESSR versus the transmit SNR P_{tot}/σ^2 .

$\lambda/2$ UPA increases slowly due to its small aperture. As a result, the gain of the MA over the FPA reaches 34.6% at $\Delta = 0.2$ rad. The sparse FPA outperforms the compact FPA owing to its larger aperture, but it is still inferior to the MA by 6.5%–18.7%, since its fixed geometry cannot adapt to the directions of the LUs and Eve, and its grating lobes may be steered toward Eve. Moreover, the per-user power allocation improves the ESSR over the equal power allocation by 2.5% at $\Delta = 0.02$ rad, where the secrecy rate of LU 1 is low and the power allocated to it can be reduced, whereas the two schemes perform similarly for large Δ .

Fig. 9 shows the ESSR versus the side length A of the movable region. The ESSR of the proposed design increases with A , since a larger region provides more spatial degrees of freedom for decorrelating the LoS components of the LUs and Eve, although the increase slows down as A grows. Even for $A = 1.5\lambda$, where the region is only slightly larger than the footprint of the $\lambda/2$ UPA, the MA improves the ESSR by 8.8% over the FPA by rearranging the antennas within the region. For $A = 6\lambda$, the gain increases to 34.3%. Although the sparse FPA also benefits from the enlarged aperture, the

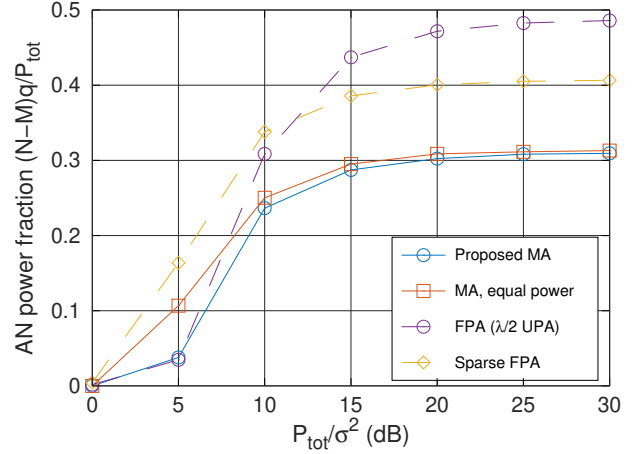


Fig. 7. Optimized AN power fraction $(N - M)q/P_{\text{tot}}$ versus the transmit SNR.

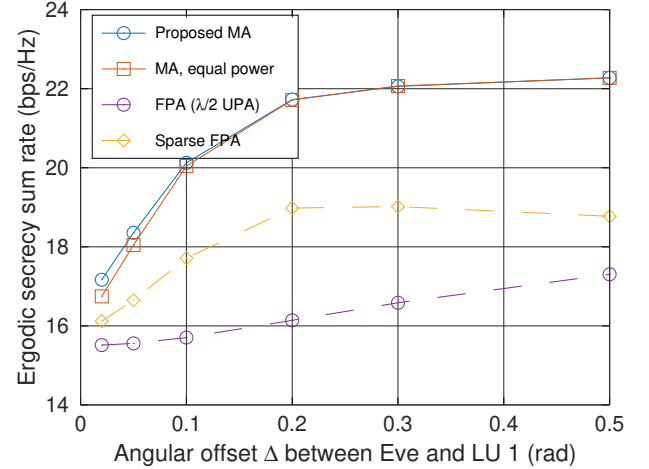
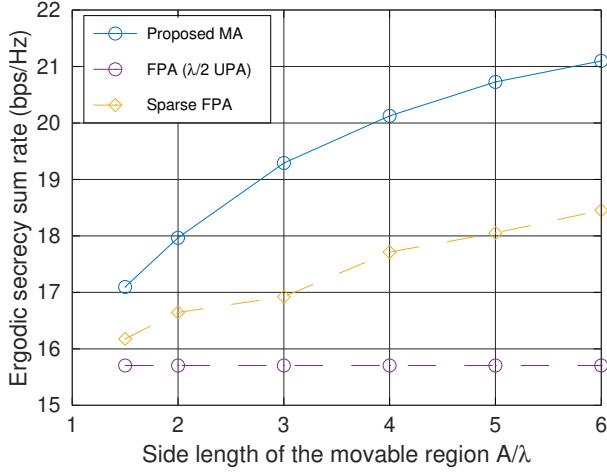
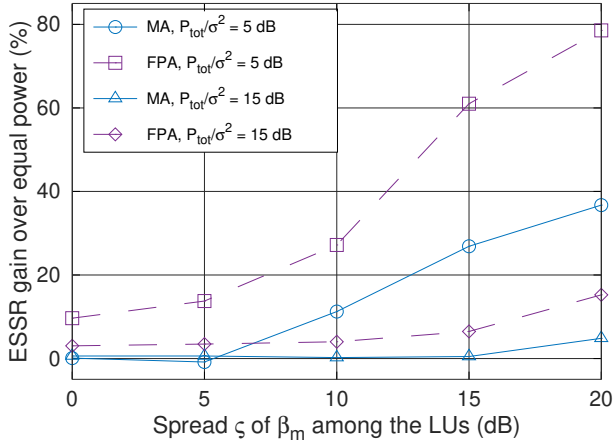


Fig. 8. ESSR versus the angular offset Δ between Eve and LU 1.

MA consistently outperforms it by 5.7%–14.8%.

F. Benefit of the Long-Term Power Allocation

In Figs. 6 and 8, the per-user power allocation brings only marginal gains over the equal power allocation, since the LUs have identical large-scale fading and the optimized antenna positions balance their secrecy conditions. To examine the benefit of the long-term power allocation for heterogeneous LUs, Fig. 10 shows the ESSR gain of the proposed per-user power allocation over the equal power allocation versus the spread ς of the large-scale fading coefficients, where $[\beta_1, \beta_2, \beta_3] = [0, -\varsigma/2, -\varsigma]$ dB. It is observed that the gain increases with ς and is more pronounced at low SNR. For example, for $\varsigma = 20$ dB, the gains for the MA and the FPA are 36.7% and 78.6% at $P_{\text{tot}}/\sigma^2 = 5$ dB, and 4.8% and 15.2% at $P_{\text{tot}}/\sigma^2 = 15$ dB, respectively. This is because the secrecy water-filling in Proposition 3 allocates the power according to the effective channel gains of the LUs and Eve and switches off the LUs with non-positive secrecy rates, whereas the equal power allocation wastes power on the weak LUs. At high SNR, equal power allocation becomes nearly optimal, similar

Fig. 9. ESSR versus the side length A of the movable region.Fig. 10. ESSR gain of the long-term per-user power allocation over the equal power allocation versus the spread ζ of the large-scale fading coefficients of the LUs, where $[\beta_1, \beta_2, \beta_3] = [0, -\zeta/2, -\zeta]$ dB.

to conventional water-filling. Moreover, the gain for the FPA is larger than that for the MA, since the FPA cannot improve the secrecy conditions of the LUs that are close to Eve by antenna positioning and thus relies more on power allocation. The only exception is a slight loss of 0.8% for the MA at $P_{\text{tot}}/\sigma^2 = 5$ dB and $\zeta = 5$ dB, which is caused by the mismatch between the approximate objective, for which the per-user power allocation improves the objective value by 2.9%, and the exact ESSR. These results show that the long-term per-user power allocation is essential for heterogeneous LUs and complements the MA positioning.

G. Single-LU Case

Fig. 11 shows the single-LU case, i.e., $M = 1$. As discussed in Remark 3, the ergodic rate of the LU does not depend on the antenna positions for a given power, which implies that the MAs bring no gain in the rate-oriented design [16]. In contrast, Fig. 11(a) shows that the MAs substantially improve the ergodic secrecy rate. For example, for $\Delta = 0.1$ rad, the proposed design achieves an ergodic secrecy rate of 7.53 bps/Hz, which

is 78.2% and 15.3% higher than those of the FPA and the sparse FPA, respectively. This gain stems from the reduced LoS correlation $|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ between the LU and Eve, which simultaneously reduces the information leakage and increases the AN received by Eve according to (29) and (30). Moreover, Fig. 11(b) shows that the MAs reduce the optimized AN power fraction from 0.84 to 0.50 for $\Delta = 0.1$ rad and to about 0.28 for $\Delta \geq 0.2$ rad, i.e., more power is allocated to the confidential signal. This verifies the AN-saving effect revealed in Remark 4. For $\Delta = 0.02$ rad, the gain of the MA reduces to 6.4%, since Eve is almost aligned with the LU.

VI. CONCLUSION

This paper proposed a two-timescale secure transmission framework for an MA-enabled multiuser downlink system with a passive Eve, where the antenna positions and the long-term power allocation among the confidential signals and AN are optimized based on the statistical CSI, and ZF precoding and null-space AN are designed based on the instantaneous CSI of the LUs. By exploiting the conditional independence between the information leakage and the AN received by Eve, a closed-form approximation of the ergodic wiretap rate was derived, which shows that the antenna positions affect the ESSR only through three geometric quantities characterizing the decorrelation among the LUs, the information leakage, and the AN received by Eve. Based on this approximation, an AO algorithm was developed, where the power allocation admits a secrecy water-filling structure and the antenna positions are optimized element-wise by the MM method. Numerical results demonstrated that the proposed two-timescale design significantly outperforms the FPA-based benchmarks, enhances the secrecy performance even for a single LU, and reduces the AN power required for secure transmission. Extending the proposed framework to multiple non-colluding or colluding eavesdroppers, multi-antenna eavesdroppers, and imperfect CSI of the LUs is an interesting direction for future work.

APPENDIX A PROOF OF LEMMA 3

For brevity, let $\Phi \triangleq \Phi_{\mathbf{F}}$. We first prove (22). Replacing $(\mathbf{F}^H \mathbf{F})^{-1}$ in the unnormalized ZF direction by $\Phi^{-1} = (\mathbb{E}\{\mathbf{F}^H \mathbf{F}\})^{-1}$ yields $\bar{\mathbf{w}}_m(\mathbf{F}) \approx \mathbf{F} \Phi^{-1} \mathbf{e}_m / \|\mathbf{F} \Phi^{-1} \mathbf{e}_m\|$. Then, we approximate the mean of $|\mathbf{b}^H \mathbf{F} \Phi^{-1} \mathbf{e}_m|^2 / \|\mathbf{F} \Phi^{-1} \mathbf{e}_m\|^2$ by the ratio of the means of its numerator and denominator, in the same spirit as [19, Lemma 1]. Since $\mathbb{E}\{\mathbf{W}\} = \mathbf{0}$ and $\mathbb{E}\{\mathbf{W} \mathbf{x} \mathbf{x}^H \mathbf{W}^H\} = \|\mathbf{x}\|^2 \mathbf{I}_N$ for any deterministic $\mathbf{x}$, we have

$$\begin{aligned} \mathbb{E}\{|\mathbf{b}^H \mathbf{F} \Phi^{-1} \mathbf{e}_m|^2\} &= |\mathbf{b}^H \bar{\mathbf{F}} \Phi^{-1} \mathbf{e}_m|^2 + \|\mathbf{b}\|^2 \|\Phi^{-1} \mathbf{e}_m\|^2 \\ &= |[\Phi^{-1} \bar{\mathbf{F}}^H \mathbf{b}]_m|^2 + \|\mathbf{b}\|^2 [\Phi^{-2}]_{m,m}, \end{aligned} \quad (54)$$

$$\mathbb{E}\{\|\mathbf{F} \Phi^{-1} \mathbf{e}_m\|^2\} = \mathbf{e}_m^T \Phi^{-1} \Phi \Phi^{-1} \mathbf{e}_m = [\Phi^{-1}]_{m,m}, \quad (55)$$

which lead to (22). Next, we prove (23). Replacing $(\mathbf{F}^H \mathbf{F})^{-1}$ in $\mathbf{P}_{\mathbf{F}} \triangleq \mathbf{I}_N - \mathbf{P}_{\mathbf{F}}^\perp = \mathbf{F}(\mathbf{F}^H \mathbf{F})^{-1} \mathbf{F}^H$ by Φ^{-1} and using $\mathbb{E}\{\mathbf{W} \Phi^{-1} \mathbf{W}^H\} = \text{tr}(\Phi^{-1}) \mathbf{I}_N$ yield

$$\mathbb{E}\{\mathbf{P}_{\mathbf{F}}\} \approx \mathbf{\Pi} + \text{tr}(\Phi^{-1}) \mathbf{I}_N, \quad \mathbf{\Pi} \triangleq \bar{\mathbf{F}} \Phi^{-1} \bar{\mathbf{F}}^H, \quad (56)$$

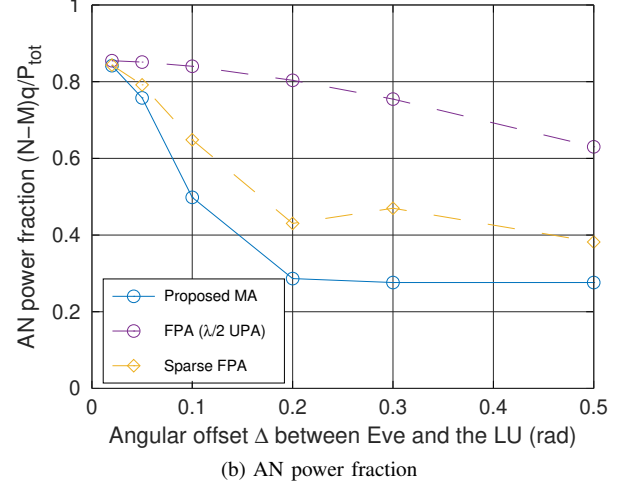
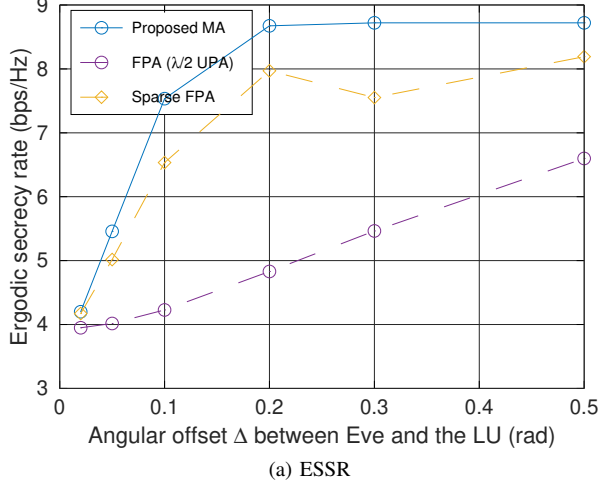


Fig. 11. Single-LU case ($M = 1$): (a) ergodic secrecy rate and (b) optimized AN power fraction versus the angular offset Δ between Eve and the LU.

where Π and $\text{tr}(\Phi^{-1})\mathbf{I}_N$ represent the parts of the signal subspace induced by the LoS and NLoS components, respectively, i.e., the NLoS component captures on average a fraction $\text{tr}(\Phi^{-1})$ of every direction. However, the directions that have already been captured by the LoS component cannot be captured again by the NLoS component. Directly using (56) counts the energy of $\mathbf{b}$ in these directions twice, and $\mathbf{b}^H(\mathbf{I}_N - \Pi - \text{tr}(\Phi^{-1})\mathbf{I}_N)\mathbf{b}$ may even become negative when $\mathbf{b}$ is strongly aligned with the column space of $\bar{\mathbf{F}}$. To avoid this double counting, we apply the NLoS-induced fraction only to the complement of Π , i.e., $\mathbb{E}\{\mathbf{P}_{\bar{\mathbf{F}}}\} \approx \Pi + \text{tr}(\Phi^{-1})(\mathbf{I}_N - \Pi)$, which yields $\mathbb{E}\{\mathbf{P}_{\bar{\mathbf{F}}}\} \approx (1 - \text{tr}(\Phi^{-1}))(\mathbf{I}_N - \Pi)$ and thus (23).

Range: Let $\mu_1, \dots, \mu_k$ denote the singular values of $\bar{\mathbf{F}}$. The nonzero eigenvalues of Π are $\mu_i^2/(\mu_i^2 + N) < 1$, and thus $\mathbf{0} \preceq \Pi \prec \mathbf{I}_N$. Moreover, $\Phi \succeq N\mathbf{I}_k$ implies $\text{tr}(\Phi^{-1}) \leq k/N < 1$. Hence, the right-hand side of (23) lies in $[0, \|\mathbf{b}\|^2]$. For (22), by the Cauchy-Schwarz inequality, $|\Phi^{-1}\bar{\mathbf{F}}^H\mathbf{b}|_m \leq \|\mathbf{b}\|^2 \mathbf{e}_m^T \Phi^{-1} \bar{\mathbf{F}}^H \bar{\mathbf{F}} \Phi^{-1} \mathbf{e}_m$, and $\|\mathbf{b}\|^2 [\Phi^{-2}]_{m,m} \leq N \|\mathbf{b}\|^2 [\Phi^{-2}]_{m,m}$. Since $\mathbf{e}_m^T \Phi^{-1} (\bar{\mathbf{F}}^H \bar{\mathbf{F}} + N\mathbf{I}_k) \Phi^{-1} \mathbf{e}_m = [\Phi^{-1}]_{m,m}$, the right-hand side of (22) lies in $[0, \|\mathbf{b}\|^2]$.

Exactness for $\bar{\mathbf{F}} = \mathbf{0}$: In this case, $\Phi = N\mathbf{I}_k$ and $\Pi = \mathbf{0}$, and the right-hand sides of (22) and (23) reduce to $\|\mathbf{b}\|^2/N$ and $\|\mathbf{b}\|^2(1 - k/N)$, respectively. On the other hand, since $\mathbf{W}$ is unitarily invariant, i.e., $\mathbf{U}\mathbf{W}$ has the same distribution as $\mathbf{W}$ for any unitary matrix $\mathbf{U}$, and $\bar{\mathbf{w}}_m(\mathbf{U}\mathbf{W}) = \mathbf{U}\bar{\mathbf{w}}_m(\mathbf{W})$, $\bar{\mathbf{w}}_m(\mathbf{W})$ is uniformly distributed on the unit sphere, and $\mathbb{E}\{\mathbf{P}_{\bar{\mathbf{W}}}\}$ commutes with every unitary matrix. Hence, $\mathbb{E}\{\|\mathbf{b}^H \bar{\mathbf{w}}_m(\mathbf{W})\|^2\} = \|\mathbf{b}\|^2/N$ and $\mathbb{E}\{\mathbf{P}_{\bar{\mathbf{W}}}\} = (1 - k/N)\mathbf{I}_N$, where the latter follows from $\text{tr}(\mathbf{P}_{\bar{\mathbf{W}}}) = N - k$.

Asymptotic exactness for $\bar{\mathbf{F}} = \varrho \bar{\mathbf{F}}_0$: As $\varrho \rightarrow \infty$, $\bar{\mathbf{F}}/\varrho \rightarrow \bar{\mathbf{F}}_0$ almost surely. Since $\bar{\mathbf{w}}_m(\cdot)$ and $\mathbf{P}_{(\cdot)}^\perp$ are invariant to the scaling of their arguments and continuous at full-column-rank matrices, and $\|\mathbf{b}^H \bar{\mathbf{w}}_m(\mathbf{F})\|^2$ and $\mathbf{b}^H \mathbf{P}_{\bar{\mathbf{F}}}^\perp \mathbf{b}$ are bounded by $\|\mathbf{b}\|^2$, the dominated convergence theorem yields $\mathbb{E}\{\|\mathbf{b}^H \bar{\mathbf{w}}_m(\mathbf{F})\|^2\} \rightarrow \|\mathbf{b}^H \bar{\mathbf{w}}_m(\bar{\mathbf{F}}_0)\|^2$ and $\mathbb{E}\{\mathbf{b}^H \mathbf{P}_{\bar{\mathbf{F}}}^\perp \mathbf{b}\} \rightarrow \mathbf{b}^H \mathbf{P}_{\bar{\mathbf{F}}_0}^\perp \mathbf{b}$. On the other hand, $\varrho^2 \Phi^{-1} \rightarrow (\bar{\mathbf{F}}_0^H \bar{\mathbf{F}}_0)^{-1}$, $\Pi \rightarrow \bar{\mathbf{F}}_0 (\bar{\mathbf{F}}_0^H \bar{\mathbf{F}}_0)^{-1} \bar{\mathbf{F}}_0^H$, and $\text{tr}(\Phi^{-1}) \rightarrow 0$. Hence, the right-hand

side of (23) converges to $\mathbf{b}^H \mathbf{P}_{\bar{\mathbf{F}}_0}^\perp \mathbf{b}$, and that of (22) converges to $|\mathbf{b}^H \bar{\mathbf{F}}_0 (\bar{\mathbf{F}}_0^H \bar{\mathbf{F}}_0)^{-1} \bar{\mathbf{F}}_0^H \mathbf{b}|^2 / |(\bar{\mathbf{F}}_0^H \bar{\mathbf{F}}_0)^{-1}|_{m,m} = |\mathbf{b}^H \bar{\mathbf{w}}_m(\bar{\mathbf{F}}_0)|^2$, since the second term in the numerator of (22) decays as $\mathcal{O}(\varrho^{-4})$ whereas the denominator decays as $\mathcal{O}(\varrho^{-2})$. This completes the proof.

APPENDIX B DERIVATIVES OF ω_m , c_m , AND d

Let $o_m \triangleq [\Omega^2]_{m,m}$, $\tau \triangleq \text{tr}(\Omega)$, $\tilde{\mathbf{v}} \triangleq \Omega \mathbf{v}$, and $\varsigma \triangleq \mathbf{v}^H \Omega \mathbf{v}$. Since Ω is Hermitian, (47) can be rewritten as $\dot{\Omega} = -(\Omega \dot{\mathbf{I}}_n)(\Omega \mathbf{I}_n)^H - (\Omega \mathbf{I}_n)(\Omega \dot{\mathbf{I}}_n)^H$. Hence, the derivatives of ω_m , o_m , and τ w.r.t. $u \in \{x_n, y_n\}$ are given by

$$\dot{\omega}_m = [\dot{\Omega}]_{m,m} = -2\text{Re}\{[\Omega \dot{\mathbf{I}}_n]_m [\Omega \mathbf{I}_n]_m^*\}, \quad (57)$$

$$\dot{o}_m = 2\text{Re}\{[\Omega \dot{\Omega}]_{m,m}\} = -2\text{Re}\{[\Omega^2 \dot{\mathbf{I}}_n]_m [\Omega \mathbf{I}_n]_m^* + [\Omega^2 \mathbf{I}_n]_m [\Omega \dot{\mathbf{I}}_n]_m^*\}, \quad (58)$$

$$\dot{\tau} = \text{tr}(\dot{\Omega}) = -2\text{Re}\{\mathbf{I}_n^H \Omega^2 \dot{\mathbf{I}}_n\}, \quad (59)$$

where (58) follows from $[\dot{\Omega} \Omega]_{m,m} = [\Omega \dot{\Omega}]_{m,m}^*$. For c_m in (24), the derivative of $\tilde{\mathbf{v}} = \Omega \mathbf{v}$ is

$$\dot{\tilde{\mathbf{v}}} = \dot{\Omega} \mathbf{v} + \Omega \dot{\mathbf{v}} = -\Omega \dot{\mathbf{I}}_n (\mathbf{I}_n^H \tilde{\mathbf{v}}) - \Omega \mathbf{I}_n (\dot{\mathbf{I}}_n^H \tilde{\mathbf{v}}) + \Omega \dot{\mathbf{v}}, \quad (60)$$

and thus

$$\dot{c}_m = \frac{2\text{Re}\{[\tilde{\mathbf{v}}]_m^* [\dot{\tilde{\mathbf{v}}}]_m\} + N\dot{o}_m - c_m \dot{\omega}_m}{\omega_m}. \quad (61)$$

For d in (25), since $\mathbf{v}^H \dot{\Omega} \mathbf{v} = -2\text{Re}\{(\tilde{\mathbf{v}}^H \dot{\mathbf{I}}_n)(\mathbf{I}_n^H \tilde{\mathbf{v}})\}$, we have

$$\dot{\varsigma} = 2\text{Re}\{\tilde{\mathbf{v}}^H \dot{\mathbf{v}}\} - 2\text{Re}\{(\tilde{\mathbf{v}}^H \dot{\mathbf{I}}_n)(\mathbf{I}_n^H \tilde{\mathbf{v}})\}, \quad (62)$$

$$\dot{d} = -\dot{\varsigma}(1 - \tau) - (N - \varsigma)\dot{\tau}. \quad (63)$$

Substituting $\dot{\eta}_m = -\eta_m \dot{\omega}_m / \omega_m$, (61), and (63) into (48) yields $\nabla_{\mathbf{t}_n} F$. All the quantities in (57)–(63) are obtained from matrix-vector products with Ω , which require $\mathcal{O}(M^2)$ operations.

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